Strict Higher Categories Model Spaces in Every Dimension

A self-contained account of the theorem of Ara and Maltsiniotis

2026

Abstract

For every $1 \leq n \leq \infty$, the category of small strict globular n-categories admits a proper combinatorial model structure that is Quillen equivalent to the standard model category of simplicial sets. Weak equivalences and fibrations are detected by the twice-extended Street nerve $\mathrm{Ex}^2 N_n$. The Quillen equivalence is the adjunction $c_n \mathrm{Sd}^2 \dashv \mathrm{Ex}^2 N_n$. This proves the higher-dimensional Thomason model–structure conjecture of Ara and Maltsiniotis.

This writeup gives a complete proof. It is addressed to a reader who has seen the basic language of categories and topological spaces, but who may never have met orientals, the Gray tensor product, or the small-object argument. Every technical step is spelled out, and every result borrowed from the literature is cited at the exact point of use. The path from a strict n-category to a space follows three machines: a positive relative prism that makes a strict cylinder stationary on an old subcategory, a family of fibre and sieve calculations that let covers be glued, and the small-object argument that assembles everything into a model structure.

Contents

1 What the theorem says and why it matters **2**
- 1.1 The question . . . 2
- 1.2 The main theorem . . . 3
- 1.3 How the proof goes: three machines . . . 4
- 1.4 Relation to earlier work . . . 5
- 1.5 Organization and conventions . . . 5

2 Foundations **6**
- 2.1 Strict globular categories . . . 6
- 2.2 Orientals and the Street nerve . . . 6
- 2.3 Subdivision, extension, and the prescribed classes . . . 7
- 2.4 Augmented directed complexes and Steiner's functor . . . 7
- 2.5 The Gray tensor product . . . 8
- 2.6 Right cylinders and truncation . . . 8
- 2.7 A strict right cylinder gives a nerve homotopy . . . 9
- 2.8 Filtered colimits and local presentability . . . 10
- 2.9 The simplicial gluing lemma . . . 11

3 The positive relative prism **11**
- 3.1 The relative prism lemma . . . 12
- 3.2 Gluing a stationary cylinder through a pushout . . . 14

4 Labelled pieces of algebraic pushouts **14**
4.1 Fibers preserve colimits . . . 15
4.2 Oriental segments and poset pieces . . . 15
4.3 Fibers of a labelled pushout . . . 16
4.4 Nerves of two-piece covers . . . 17

5 Couniversality and the unit on posets **18**
5.1 One greatest new element . . . 18
5.2 Small finite posets: induction through two-piece covers . . . 19
5.3 The unit on posets . . . 20

6 Comparison for subdivided attachments **20**
6.1 The attachment comparison . . . 21
6.2 The twice-subdivided boundary and horn . . . 22
6.3 Horn attachments are acyclic . . . 23

7 The model structure and the Quillen equivalence **23**
7.1 The simplicial inputs . . . 24
7.2 Transfinite horn attachments . . . 24
7.3 The model structure . . . 24
7.4 Properness . . . 25
7.5 The unit after two subdivisions . . . 26
7.6 The Quillen equivalence . . . 26

A Local finite presentability of nCat **27**
A.1 Signature and equations . . . 27
A.2 Defined terms and presentations . . . 28
A.3 Colimits, filtered colimits, finite presentations . . . 29

1 What the theorem says and why it matters

1.1 The question

Topology has a useful trick. A space X is often hard to study directly. But spaces can be approximated by combinatorial models: cell complexes, simplicial sets, simplicial groups, and so on. Two spaces are considered equivalent if they can be deformed into each other (they are *homotopy equivalent*). The goal of *homotopy theory* is to understand spaces up to this deformation. The collective insight of the last fifty years is that a very wide range of combinatorial objects carry their own internal notion of *weak equivalence* (a notion of "these two objects describe the same space"), and that this notion assembles into a rich algebraic structure called a *model category*.

The question we answer here is a refinement of a landmark result of Thomason [16]. Call a functor between small categories a *Thomason equivalence* if the ordinary nerve construction turns it into a weak homotopy equivalence of simplicial sets. Thomason proved that the collection of small categories carries a model structure whose weak equivalences are exactly these, and that this model structure is Quillen equivalent to the model structure on simplicial sets. In plain terms, categories are just as good a model of spaces as simplicial sets. The bridge is provided by the composite $c_1\mathrm{Sd}^2 \dashv \mathrm{Ex}^2 N_1$, where N_1 is the ordinary nerve, c_1 is its left adjoint (the "categorification" functor),

and $\mathrm{Sd} \dashv \mathrm{Ex}$ is the classical subdivision–extension adjunction of simplicial sets. Two subdivisions are needed: one is not enough.

Strict higher categories generalise categories in another direction. A strict globular n-category has collections of k-cells for $k = 0, 1, \dots, n$. Objects are 0-cells; a 1-cell $x \to y$ is an arrow; a 2-cell $\alpha\colon x \Rightarrow y$ sits between parallel 1-cells; a 3-cell sits between parallel 2-cells; and so on. Composition is strict: it is associative on the nose and obeys unit and interchange laws exactly. When $n = 1$ we recover ordinary categories, when $n = \infty$ we get strict globular ω-categories, and the "strict" in the name is deliberate: positive-dimensional cells need not be invertible.

The Street nerve N_n makes strict n-categories into simplicial sets, and it has a left adjoint c_n given by a colimit of *orientals*. The higher–dimensional Thomason conjecture of Ara and Maltsiniotis asks whether the composite $L_n = c_n\mathrm{Sd}^2$, together with its right adjoint $R_n = \mathrm{Ex}^2 N_n$, transfers the Kan–Quillen model structure from sSet to nCat, and whether the resulting model structure is Quillen equivalent to sSet via (L_n, R_n). Ordinary Thomason is the case $n = 1$.

1.2 The main theorem

We use the following fixed notation.

nCat is the category of small strict globular n-categories and strict functors; $\omega\mathrm{Cat} = 1\mathrm{Cat}_\omega$ is the case $n = \infty$; sSet is the category of simplicial sets.

$\mathcal{O}_m$ is the mth *oriental* of Street. Its defining property is that maps from $\mathcal{O}_m$ to a strict ω-category X determine what is called an *m-simplex* of X, with all expected face and degeneracy compatibilities. The orientals assemble into a cosimplicial object $\mathcal{O}_\bullet\colon \Delta \to \omega\mathrm{Cat}$.

For $1 \le n \le \infty$, the *Street nerve* is the simplicial set

$$N_n X\colon [m] \longmapsto \mathrm{Hom}_{\omega\mathrm{Cat}}(\mathcal{O}_m, X) = \mathrm{Hom}_{n\mathrm{Cat}}(\tau_n \mathcal{O}_m, X),$$

where $\tau_n\colon \omega\mathrm{Cat} \to n\mathrm{Cat}$ is the left adjoint to inclusion (retaining cells only in dimensions $\le n$). The left adjoint of N_n is the functor $c_n\colon \mathrm{sSet} \to n\mathrm{Cat}$ defined degreewise by the colimit of orientals $c_n K = \mathrm{colim}_{\Delta[k]\to K} \tau_n \mathcal{O}_k$.

$\mathrm{Sd} \dashv \mathrm{Ex}$ is the classical subdivision adjunction on sSet: $\mathrm{Sd}\Delta[m]$ is the ordered nerve of the poset of nonempty subsets of $\{0, \dots, m\}$, and $(\mathrm{Ex}K)_m = \mathrm{Hom}_{\mathrm{sSet}}(\mathrm{Sd}\Delta[m], K)$. There are natural maps $\ell\colon \mathrm{Sd} \to \mathrm{id}$ and $j\colon \mathrm{id} \to \mathrm{Ex}$, both objectwise weak homotopy equivalences; both statements are classical.

Set

$$L_n = c_n\mathrm{Sd}^2\colon \mathrm{sSet} \longrightarrow n\mathrm{Cat}, \qquad R_n = \mathrm{Ex}^2 N_n\colon n\mathrm{Cat} \longrightarrow \mathrm{sSet}.$$

The composite $L_n \dashv R_n$ is again an adjunction.

Let W_{KQ} denote the class of weak homotopy equivalences in sSet and $\mathrm{Fib}_{\mathrm{KQ}}$ the class of Kan fibrations. Define three classes of morphisms in nCat:

$$W_n = R_n^{-1}(W_{\mathrm{KQ}}) = N_n^{-1}(W_{\mathrm{KQ}}), \quad F_n = R_n^{-1}(\mathrm{Fib}_{\mathrm{KQ}}), \quad C_n = {}^{\perp}(F_n \cap W_n). \tag{1.1}$$

Here ${}^{\perp}S$ denotes the left lifting class of S: the class of maps that lift on the left against every map in S. The two descriptions of W_n agree because Ex preserves and reflects weak equivalences (a consequence of j being an objectwise weak equivalence; see §2).

Set

$$I = \{\partial\Delta[m] \hookrightarrow \Delta[m] : m \ge 0\},$$
$$J = \{\Lambda^k[m] \hookrightarrow \Delta[m] : m \ge 1,\ 0 \le k \le m\}.$$

Theorem 1.1 (Main theorem)**.** *For every $n \in \{1, 2, 3, \dots\} \cup \{\infty\}$, the three classes (C_n, W_n, F_n) in (1.1) form a proper combinatorial model structure on nCat. It is cofibrantly generated by (L_nI, L_nJ). The adjunction*

$$L_n \colon \mathrm{sSet}_{\mathrm{KQ}} \rightleftarrows (n\mathrm{Cat}, C_n, W_n, F_n) \colon R_n$$

is a Quillen equivalence.

In words: nCat is a model category in which a map is a weak equivalence iff its twice-extended Street nerve is a weak homotopy equivalence, a fibration iff its twice-extended Street nerve is a Kan fibration, and a cofibration iff it has the left lifting property against the simultaneous weak equivalences and fibrations. The pair (L_n, R_n) lifts the standard simplicial model structure to nCat on the nose, and it is an equivalence of homotopy theories.

1.3 How the proof goes: three machines

A reader meeting the theorem for the first time may wonder why it is not obvious. There are two obstacles that do not occur in dimension one.

The subdivision obstacle. One subdivision Sd is not enough to make the natural map $\mathrm{Sd}K \to N_1 c_1 \mathrm{Sd}K$ a weak equivalence; a second subdivision fixes the problem by iterating the direction of the subdivision. The same phenomenon occurs here, and we defer it entirely: the subdivision facts are inputs, and the meaningful work happens after subdivision.

The pushout obstacle. To build a model structure by transfer from sSet, one needs to know that the small-object construction behaves well. Concretely, for every map d in the generating set J, one needs that any pushout of L_nd is a weak equivalence in nCat. But L_nd is a map of the form $U_A \to U_B$, where $U_E = c_nNE$ is the strict n-category attached to a poset E and N is the ordinary nerve. So the question becomes: when is a pushout

$$X \longrightarrow X \amalg_{U_A} U_B$$

a weak equivalence, for every attaching functor $U_A \to X$?

Thomason solved this in dimension one by identifying a class of poset inclusions — the *Dwyer maps* — with the property that their every cobase change is a weak equivalence. The heart of the present proof is the discovery that this class admits a clean synthesis in every higher dimension.

Machine 1: the positive relative prism. Fix a downward-closed poset inclusion $A \subset E$ (a "sieve"). Suppose $f, g \colon E \to E$ are monotone with $f \leq g$, with f and g agreeing on A, and with g constant on the complement $E \setminus A$. Then there is an explicit *strict right Gray cylinder*

$$P_{f,g} \colon U_E \otimes D_1 \longrightarrow U_E$$

whose two endpoints are $U(f)$ and $U(g)$, and which is *stationary* on U_A: at every parameter value it acts on U_A by the common restriction $f|_A = g|_A$. The construction is a simplexwise chain computation using the Alexander–Whitney corner, and it survives descent to every finite dimension by an internal-hom calculation. The stationarity is what makes everything glue.

Machine 2: covers. Given a sieve $A \subseteq B$ whose inclusion into its upward closure Z has a right-adjoint retraction $r \colon Z \to A$, one uses the prism to contract $A \hookrightarrow Z$ *couniversally*: for every attaching functor $U_A \to X$, the map $X \to X \amalg_{U_A} U_Z$ is a weak equivalence. Then one covers B by the two cosieves Z and $B \setminus A$, glues by simplicial left properness, and obtains the comparison between a pushout of nerves and the nerve of the pushout. Applied to the twice-subdivided boundaries and horns, this supplies the model-structure inputs.

Machine 3: transfer. With the acyclicity and invariance inputs in hand, the small-object argument runs as in any transfer proof. The subtlety is that the small-object factorisation is computed in nCat, not in sSet; this is where the requirement that all attaching functors be allowed to be arbitrary matters. The unit of the adjunction $L_n \dashv R_n$ on an arbitrary simplicial set is assembled from the poset case by gluing along boundaries, using left properness of sSet.

1.4 Relation to earlier work

Thomason [16] proves the case $n = 1$. Cisinski [7] corrects the retract argument used for properness. Ara and Maltsiniotis developed an abstract transfer theorem and proved the case $n = 2$ together with the Quillen equivalence [4]. They also state the general conjecture; Ara's 2023 manuscript [3] records the remaining condition and Guetta and Maltsiniotis [11] restate the ω-dimensional case. The positive-poset and unit conditions in all dimensions were proved earlier [5, 9]; the missing ingredient was the couniversality of a sieve with a right-adjoint retraction, which the present relative prism supplies in arbitrary target categories.

1.5 Organization and conventions

Sections 2–6 are the mathematical core. Section 2 fixes notation and proves the small foundational lemmas. Section 3 constructs the relative prism and proves the cylinder-gluing lemma. Section 4 does the fibre and sieve calculations. Section 5 proves couniversality and the unit on posets. Section 6 proves the comparison for subdivided boundary and horn attachments. Section 7 assembles the model structure, properness, and Quillen equivalence. Section A is an appendix proving local finite presentability of nCat from a signature.

Throughout, the universe is fixed and small; we write "small" for "sets", and we allow ourselves to enlarge the universe when forming categories of small objects. The symbol D_1 denotes the interval category $0 \to 1$, and D_i the free strict ω-category on one i-cell. The interval is the *right* tensor factor in every Gray-cylinder formula, per the convention of [6, Appendix B.4].

Notation at a glance

For the reader's convenience, here is a compact index of the symbols that appear most often. Detailed definitions are in the section that first introduces each item.

- nCat: small strict globular n-categories. ωCat: the case $n = \infty$.
- sSet: simplicial sets; N: ordinary nerve; N_n: Street nerve.
- $\mathcal{O}_m$: the mth oriental. D_i: the free strict ω-category on one i-cell.
- c_n: left adjoint of N_n, computed by $c_n K = \operatorname{colim}_{\Delta[m] \to K} \tau_n \mathcal{O}_m$.
- $\mathrm{Sd} \dashv \mathrm{Ex}$: subdivision-extension adjunction on simplicial sets.
- $L_n = c_n \mathrm{Sd}^2$, $R_n = \mathrm{Ex}^2 N_n$: the two sides of the Quillen equivalence.
- W_n, F_n, C_n: weak equivalences, fibrations, cofibrations.
- $U_E = c_n N E$: strict n-category attached to a poset E.
- ν: Steiner's functor from directed chain complexes to strict ω-categories.

- $M[m]$: normalized simplex complex. $G[i]$: globe complex.
- $\otimes$: Gray tensor product. D_1: interval, used as the right factor.
- $H(Y)$: right internal hom for the Gray tensor.
- $W_{\mathrm{KQ}}, \mathrm{Fib}_{\mathrm{KQ}}$: weak equivalences and Kan fibrations in sSet.
- I, J: generating sets for the simplicial model structure.

2 Foundations

The pushout argument rests on four foundations: strict categories give a concrete presentation of the Gray tensor product through directed chain complexes; strict right cylinders correspond to certain simplicial homotopies of nerves; strict categories are locally finitely presentable, so filtered colimits are well behaved; and certain pushouts of simplicial sets are weak equivalences whenever one leg is monomorphic. We fix notation and prove the small local statements here, deferring only the presentability proof to Section A.

2.1 Strict globular categories

We fix a universe. A *globular set* is a sequence $X_0, X_1, X_2, \dots$ of sets with maps $s, t\colon X_{k+1} \to X_k$ (source and target) satisfying $ss = st$ and $ts = tt$, where the leftmost letter is the inner face. A *strict globular ω-category* is a globular set together with compositions $\circ_k$ along k-matching boundaries (for $k \geq 0$ and $x, y \in X_{k+1}$ with $sx = ty$), identities $e_k\colon X_k \to X_{k+1}$, satisfying the strict associativity, unit, and interchange laws. Positive-dimensional cells are not required to be invertible. A *strict globular n-category* for finite n is a strict ω-category with exactly one cell in each dimension $> n$. We write $n\mathrm{Cat}$ for the category of small strict n-categories and strict functors, and $\omega\mathrm{Cat} = 1\mathrm{Cat}_\omega$ (the case $n = \infty$).

The inclusion $n\mathrm{Cat} \hookrightarrow \omega\mathrm{Cat}$ has a left adjoint $\tau_n = \tau_n$. Explicitly, $\tau_n X$ retains the cells of X in dimensions below n; its n-cells are the n-cells of X modulo the congruence closed under every defined composition and generated by

$$s_n(z) = t_n(z) \qquad (z \text{ an } (n+1)\text{-cell of } X).$$

Generating pairs are parallel n-cells, so their identifications leave every lower matching boundary unchanged; compositions descend. Cells above n become identities. Every strict functor from X to an n-category annihilates these relations, proving the reflection property. Set $\tau_\infty = \mathrm{id}$.

Write D_k for the free strict ω-category on one k-cell with all its globular faces. Thus $\mathrm{Hom}_{\omega\mathrm{Cat}}(D_k, X)$ is the set of k-cells of X, D_0 is terminal, and D_1 is the category $0 \to 1$.

2.2 Orientals and the Street nerve

The mth *oriental* $\mathcal{O}_m$ is a strict ω-category freely generated by the m-simplex. The cosimplicial structure maps $[-]\colon \Delta \to \omega\mathrm{Cat}$ encode faces and degeneracies by applying a monotone map of ordinals $\alpha\colon [q] \to [m]$ to the generating chain complex; this is standard, and we take it from Street [15]. For $1 \leq n \leq \infty$ the *Street nerve*

$$N_n X\colon [m] \longmapsto \mathrm{Hom}_{\omega\mathrm{Cat}}(\mathcal{O}_m, X) = \mathrm{Hom}_{n\mathrm{Cat}}(\tau_n \mathcal{O}_m, X)$$

is a simplicial set. The two formulas agree because maps out of $\mathcal{O}_m$ into an n-category annihilate the congruence defining $\tau_n\mathcal{O}_m$. The left adjoint $c_n\colon \mathrm{sSet} \to n\mathrm{Cat}$ is computed by the canonical colimit

$$c_nK = \operatorname*{colim}_{(\Delta[m]\to K)\in\Delta/K} \tau_n\mathcal{O}_m \cong \operatorname*{colim}_{\Delta[m]\to K} \tau_n\mathcal{O}_m, \tag{2.1}$$

by the universal property of a colimit and the definition of the nerve.

For a poset E write NE for its ordinary nerve (the simplicial set of monotone maps $[m] \to E$), and

$$U_E = c_nNE, \qquad U_E^\infty = c_\infty NE,$$

so $U_E = \tau_n U_E^\infty$. For a monotone map $f\colon E \to E'$, write $U(f) = c_nN(f)$ and $U^\infty(f) = c_\infty N(f)$. On ordinary categories N_1 agrees with N_∞: a map from an oriental is determined by its successive edges, because the triangular cells impose the composition equations. In particular $N_\infty D_1 = \Delta[1]$.

2.3 Subdivision, extension, and the prescribed classes

The *subdivision* functor $\mathrm{Sd}\colon \mathrm{sSet} \to \mathrm{sSet}$ sends $\Delta[m]$ to the ordered nerve of the poset of nonempty subsets of $\{0,\dots,m\}$; it is a left adjoint, with right adjoint $\mathrm{Ex}\colon \mathrm{sSet} \to \mathrm{sSet}$ defined by $(\mathrm{Ex}K)_m = \mathrm{Hom}_{\mathrm{sSet}}(\mathrm{Sd}\Delta[m], K)$. The *last-vertex transformation* $\ell\colon \mathrm{Sd} \to \mathrm{id}$ is objectwise a weak homotopy equivalence; its mate $j\colon \mathrm{id} \to \mathrm{Ex}$ is also objectwise a weak homotopy equivalence. Both facts are classical [4, Theorem 4.5], [10, III.4.6]. Naturality of ℓ and j together with formal two-out-of-three shows that Sd and Ex preserve and reflect weak homotopy equivalences.

Set $L_n = c_n\mathrm{Sd}^2$ and $R_n = \mathrm{Ex}^2N_n$. They are adjoint by composition of adjunctions. Let W_{KQ} be the class of weak homotopy equivalences and $\mathrm{Fib}_{\mathrm{KQ}}$ the class of Kan fibrations in sSet. Define

$$W_n = R_n^{-1}(W_{\mathrm{KQ}}) = N_n^{-1}(W_{\mathrm{KQ}}), \quad F_n = R_n^{-1}(\mathrm{Fib}_{\mathrm{KQ}}), \quad C_n = {}^{\perp}(F_n \cap W_n).$$

The two descriptions of W_n agree because Ex^2 preserves and reflects W_{KQ}. Set $I = \{\partial\Delta[m] \hookrightarrow \Delta[m] : m \geq 0\}$ and $J = \{\Lambda^k[m] \hookrightarrow \Delta[m] : m \geq 1,\ 0 \leq k \leq m\}$.

2.4 Augmented directed complexes and Steiner's functor

An *augmented directed complex* is a nonnegative chain complex K of abelian groups with augmentation $\epsilon\colon K_0 \to \mathbb{Z}$ and a specified additive submonoid $K_p^+ \subset K_p$ in each degree. Morphisms are chain maps preserving augmentation and submonoids. All complexes below have a specified free basis; the positive submonoid is generated by the basis. For each basis element write $dc = d^+c - d^-c$ with disjoint positive supports. A basis is *unital* when $\epsilon((d^-)^p b) = \epsilon((d^+)^p b) = 1$ for each basis element b in degree p. A basis is *strongly loop-free* when the relations

$$a < b \quad \text{if } a \text{ occurs in } d^-b, \qquad b < a \quad \text{if } a \text{ occurs in } d^+b$$

are contained in a strict partial order on the whole basis. A complex with such a basis is a *strong Steiner complex*.

Steiner's functor ν sends augmented directed complexes to strict ω-categories. A p-cell of νK is a table

$$(x_0^-, x_0^+ \mid x_1^-, x_1^+ \mid \cdots \mid x_p^-, x_p^+), \quad x_r^\pm \in K_r^+,$$

such that

$$\epsilon x_0^- = \epsilon x_0^+ = 1, \quad dx_r^- = dx_r^+ = x_{r-1}^+ - x_{r-1}^- \ \ (1 \leq r \leq p), \quad x_p^- = x_p^+.$$

An identity appends a zero pair. A positive augmentation-preserving chain map acts entrywise, giving a strict functor. For a basis element b in degree p, its *atom* has top pair (b, b) and entries

$\langle b\rangle_r^{\pm} = (d^{\pm})^{p-r}b$: reduce the boundary by alternating positive or negative parts. For a strong Steiner complex these tables are cells, and νK is freely generated as a polygraph by these atoms; maps out of νK are determined by their values on them [13, Definitions 2.6, 2.8, Theorem 6.1].

We use two concrete complexes.

(i) The normalized simplex complex $M[m]$. Its degree-p basis is the set of strictly increasing tuples $[i_0, \dots, i_p]$ in $[m] = \{0, \dots, m\}$, with $d = \sum_j (-1)^j (\text{delete } i_j)$ for $p > 0$ and $\epsilon[i] = 1$. A monotone map of ordinals acts by applying it to the tuple; a tuple with a repeated entry becomes zero. It is strong Steiner and $\nu M[m] = \mathcal{O}_m$. The atom of a nonempty tuple has its first and last vertices as its 0-source and 0-target.

(ii) The globe complex $G[i]$. It has one top basis element in degree i and two basis elements (source and target faces) in each degree below i. Positive-degree differentials are target minus source; vertices have augmentation one. It is strong Steiner and $\nu G[i] = D_i$. For $i = 0$ there is one vertex.

Both facts are contained in [13, Examples 3.8–3.10, Example 4.7, Theorem 5.11], and the oriental identification and cosimplicial action, including repeated entries, in [14, Theorem 3.2, §4].

The tensor of augmented directed complexes is the ordinary nonnegative chain tensor with separate augmentation:

$$d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy, \qquad \epsilon(x \otimes y) = \epsilon(x)\epsilon(y) \quad (|x| = |y| = 0). \tag{2.2}$$

Its positive submonoid is generated by tensors of positive elements. The tensor product of two strong Steiner bases (as sets) is the positive basis of the tensor complex.

2.5 The Gray tensor product

The category $\omega\mathrm{Cat}$ (and $n\mathrm{Cat}$ for finite n) carries the *biclosed Gray tensor product* $\otimes\colon \omega\mathrm{Cat} \times \omega\mathrm{Cat} \to \omega\mathrm{Cat}$, which we take from Ara and Maltsiniotis [6, Theorem A.15 and §A.16]. It has unit D_0, preserves small colimits separately in each variable, and admits natural identifications

$$\nu K \otimes \nu L \;\cong\; \nu(K \otimes L) \qquad (K, L \text{ strong Steiner}), \tag{2.3}$$

natural in every morphism between strong Steiner complexes, including ones sending basis elements to sums or zero. This is the completed extension theorem; the general extension is not being inferred from the incomplete proof of [13, Theorem 7.3]. Colimit preservation is a property of the Gray tensor, not of ν.

It follows that $D_i \otimes D_1$ is generated by the atoms indexed by pairs of globe basis elements, of dimension the sum. The same holds for $\mathcal{O}_m \otimes D_1$. Globe face inclusions and simplex face inclusions identify the corresponding based subcomplexes: the positive and negative boundaries of a basis element inside the subcomplex are unchanged by inclusion, so the atom agrees.

Throughout, the interval is the *right* tensor factor. Write e_0, e_1 for the degree-zero basis elements of $M[1]$ and u for its degree-one element, so $du = e_1 - e_0$. Write $e_a\colon D_0 \to D_1$ for the endpoint functors and $\epsilon\colon D_1 \to D_0$ for the unique functor. The *projection* is $\pi_X = \mathrm{id}_X \otimes \epsilon\colon X \otimes D_1 \to X \otimes D_0 \cong X$. No interchange of factors is used.

2.6 Right cylinders and truncation

Biclosedness supplies an $\omega\mathrm{Cat}$-object $H(Y)$ and natural bijections

$$\mathrm{Hom}_{\omega\mathrm{Cat}}(X, H(Y)) \cong \mathrm{Hom}_{\omega\mathrm{Cat}}(X \otimes D_1, Y). \tag{2.4}$$

In the notation of [6, §A.18], $H(Y) = \mathrm{Hom}_{\mathrm{oplax}}(D_1, Y)$; formula (2.4) fixes the convention. Evaluation at e_0, e_1 and precomposition with π_X give, respectively, the endpoints and constant cylinders,

transposing functors of one variable. To extend cylinders through pushouts we use transposes into $H(Y)$. The next lemma shows that this stays within $n\mathrm{Cat}$ when Y is an n-category.

Lemma 2.1 (Internal hom and reflection)**.** *Let $n < \infty$ and let Y be a strict n-category. Then $H(Y)$ is a strict n-category. Consequently the reflection unit $X \to \tau_n X$ induces a natural bijection*

$$\mathrm{Hom}_{\omega\mathrm{Cat}}((\tau_n X) \otimes D_1, Y) \xrightarrow{\cong} \mathrm{Hom}_{\omega\mathrm{Cat}}(X \otimes D_1, Y). \tag{2.5}$$

This bijection respects endpoints and restriction along functors in the first variable. For $n = \infty$ the same assertion holds with $\tau_n = \mathrm{id}$.

Proof. An i-cell of $H(Y)$ is a functor $P\colon D_i \otimes D_1 \to Y$. Let $s\colon D_{i-1} \to D_i$ be the source inclusion and $p\colon D_i \to D_{i-1}$ the map sending the top i-cell to the identity of the top $(i-1)$-cell. Then $ps = \mathrm{id}$, and $q = sp$ collapses D_i onto its source. Under (2.4), the identity of the source of P is $P(q \otimes \mathrm{id})$. We show that $P = P(q \otimes \mathrm{id})$ when $i > n$.

The atoms of $D_i \otimes D_1$ outside $D_{i-1} \otimes D_1$ come from pairs: top i-generator paired with e_0, e_1, or u; target $(i-1)$-generator paired with e_0, e_1, or u. Their dimensions are respectively $i, i, i+1$ and $i-1, i-1, i$. If $i > n+1$, all exceed n. If $i = n+1$, only the target n-generator paired with e_0 or e_1 has dimension $\le n$. Either endpoint, P sends the top $(n+1)$-cell to an identity, so its source and target n-cells have equal images; these are the two comparisons needed to match P and $P(q \otimes \mathrm{id})$ on the remaining degree-n atoms. All atoms in the source face are fixed by $q \otimes \mathrm{id}$.

The two functors therefore agree on all atoms of dimension $\le n$. They also agree on higher atoms: in an n-category the image of a cell of dimension $> n$ is the iterated identity on its n-source, which is already determined. Generation by atoms proves $P = P(q \otimes \mathrm{id})$. Thus every cell of $H(Y)$ above n is an identity, so $H(Y)$ is an n-category.

Now transpose a cylinder to a functor $X \to H(Y)$. Since $H(Y)$ is an n-category, it factors uniquely through $\tau_n X$. Transposing back proves (2.5). Naturality of reflection and of the adjunction prove the restriction assertion; endpoint assertions follow by naturality of evaluation or by uniqueness of the factorization into Y. □

Remark 2.2. Even for X, Y in $n\mathrm{Cat}$, the expression $X \otimes D_1$ denotes their tensor in $\omega\mathrm{Cat}$ and may have cells above n. A map from this tensor into Y is nevertheless represented by a map $X \to H(Y)$ *in* $n\mathrm{Cat}$. This is what permits stationary cylinders to be glued by arbitrary pushouts in $n\mathrm{Cat}$.

2.7 A strict right cylinder gives a nerve homotopy

The next lemma lets a strict cylinder $P\colon X \otimes D_1 \to Y$ be turned into an ordinary simplicial homotopy of Street nerves. It uses the Alexander–Whitney diagonal. The statement is due to Ara [3, §§5.2.2–5.2.6, especially Proposition 5.2.3]; we include a proof for completeness.

Lemma 2.3 (Right Gray cylinder gives a nerve homotopy)**.** *There is a natural simplicial map*

$$J_X\colon N_\infty X \times \Delta[1] \longrightarrow N_\infty(X \otimes D_1)$$

whose restriction at the endpoint $a \in \{0,1\}$ is $N_\infty(\mathrm{id}_X \otimes e_a)$, and for which $N_\infty(\pi_X)\, J_X$ is the projection onto $N_\infty X$. Consequently a strict functor $P\colon X \otimes D_1 \to Y$ induces a simplicial homotopy between the nerves of its endpoint functors. If a functor $a\colon A \to X$ satisfies $P(a \otimes \mathrm{id}) = h\pi_A$ for some $h\colon A \to Y$, then the homotopy restricts along $N_\infty a$ to the constant homotopy of $N_\infty h$. The same applies to N_n when X, Y are n-categories.

Proof. For each m, define the Alexander–Whitney diagonal [10, IV.2, Remark 2.5]

$$\mathrm{AW}_m[i_0,\dots,i_p]=\sum_{a=0}^{p}[i_0,\dots,i_a]\otimes[i_a,\dots,i_p]$$

as a map $M[m]\to M[m]\otimes M[m]$. It is a positive chain map. In d of the right side, the term obtained by deleting a vertex strictly before or after the shared vertex matches a term in $\mathrm{AW}_m\,d$; the term deleting the final vertex of the first factor at cut a and the term deleting the initial vertex of the second factor at cut $a-1$ are equal up to sign, hence cancel by the tensor sign in (2.2). Every summand is positive. On vertices the augmentation is one; applying the counit to the first or second factor selects, respectively, only the cut $a=0$ or $a=p$. Thus both counits give the identity.

The diagonal is natural for every monotone ordinal map. For a strictly increasing image this is term by term. If the image has equalities at positions $j,j+1$, then every cut $a\le j$ yields a degenerate second factor and every cut $a\ge j+1$ yields a degenerate first factor; both sides of naturality are zero after normalization. This verifies faces and degeneracies. Only this cosimplicial naturality is claimed.

Apply ν and (2.3) to get natural functors $\delta_m\colon\mathcal{O}_m\to\mathcal{O}_m\otimes\mathcal{O}_m$. For an m-simplex $F\colon\mathcal{O}_m\to X$ and monotone $\theta\colon[m]\to[1]$ put

$$J_X(F,\theta)=(F\otimes\mathcal{O}_\theta)\delta_m,$$

where $\mathcal{O}_\theta\colon\mathcal{O}_m\to\mathcal{O}_1=D_1$ is the cosimplicial structure map. The naturality just verified shows that J_X is simplicial. If θ is constant at a, the map factors through the right counit and the endpoint e_a, giving $J_X(-,a)=N_\infty(\mathrm{id}_X\otimes e_a)$. Applying π_X also yields the right counit, so $N_\infty(\pi_X)J_X$ is the projection. Postcomposing with $N_\infty P$ proves the homotopy assertion. Naturality in X and the projection formula prove stationarity. □

2.8 Filtered colimits and local presentability

Strict n-categories admit a simple filtered-colimit theory. The argument is deferred to Section A, and we only record the consequences here.

Lemma 2.4 (Filtered colimits)**.** *For every $1\le n\le\infty$ the category n*Cat *is locally finitely presentable; filtered colimits in n*Cat *are computed on cells. Each $\tau_n\mathcal{O}_m$ is finitely presentable in n*Cat*. Consequently N_n, and $\mathrm{Ex}^r N_n$ for every finite $r\ge 0$, preserve filtered colimits. Weak homotopy equivalences of simplicial sets are closed under filtered colimits of arrows, as are the maps f of n*Cat *for which $N_n f$ is a weak homotopy equivalence. No monomorphism hypothesis on transition maps is required.*

Proof. The first two claims are proved in Proposition A.3. For the third: an oriental has a finite polygraphic presentation with one generator per nonempty strictly increasing tuple. A map from $\mathcal{O}_m$ into a filtered colimit lifts to one stage, and equality of two such maps is detected there; the same argument applies to $\tau_n\mathcal{O}_m$. Hence $\mathcal{O}_m$ and $\tau_n\mathcal{O}_m$ are finitely presentable, and the colimits in (2.1) display N_n as preserving filtered colimits degreewise. Indeed $(\mathrm{Ex}K)_m=\mathrm{Hom}(\mathrm{Sd}\Delta[m],K)$ with $\mathrm{Sd}\Delta[m]$ finite, so Ex also preserves filtered colimits.

Now the weak-equivalence claim. Let $Q=\mathrm{Ex}^\infty$. It preserves filtered colimits because Ex does and colimits commute with colimits. A filtered colimit of Kan complexes is Kan: any horn has finitely many nondegenerate simplices and its equations occur at one stage. For a filtered diagram of Kan complexes K_i, choose a vertex x of $\operatorname{colim}K_i$ represented by $x_i\in K_i$. The coslice $(i\to j)$ is filtered with final projection onto the base; for each $k\ge 1$,

$$\operatorname*{colim}_{(i\to j)}\pi_k(K_j,x_j)\cong\pi_k(\operatorname*{colim}_j K_j,x),$$

because the pointed sphere $\Delta[k]/\partial\Delta[k]$ and its finite reduced cylinder represent classes and homotopies in finite data, all of which lift to a sufficiently late stage. This proves both surjectivity and injectivity of the comparison, including eventual equalities without injectivity of transitions. It follows that an objectwise weak equivalence between filtered diagrams of Kan complexes induces a weak equivalence on colimits. Combining with $K \to QK$ and two-out-of-three gives the claim for W_n, and then N_n preservation gives it for $W_n = N_n^{-1}(W_{\mathrm{KQ}})$. □

2.9 The simplicial gluing lemma

We need one properness-type statement for the Kan–Quillen model structure, phrased so that it isolates the monomorphic leg.

Lemma 2.5 (Gluing with one monomorphic leg)**.** *Consider a map of spans of simplicial sets*

$$\begin{array}{ccccc} A & \longleftarrow & C & \xrightarrow{\ i\ } & B \\ {\scriptstyle a}\downarrow & & {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle b} \\ A' & \longleftarrow & C' & \xrightarrow[\ i'\ ]{} & B'. \end{array}$$

where i and i' are monomorphisms. If a, b, c are weak equivalences, then the induced map $A \amalg_C B \to A' \amalg_{C'} B'$ is a weak equivalence. There is no condition on $C \to A$ or $C' \to A'$. Each displayed pushout computes a homotopy pushout. In particular, cobase change of a weak equivalence along a simplicial monomorphism is a weak equivalence.

Proof. Put $E = B \amalg_C C'$. By simplicial left properness $B \to E$ is a weak equivalence, as the pushout of c along the monomorphism i. Two-out-of-three gives $E \to B'$ a weak equivalence. Both $C' \to E$ and $C' \to B'$ are cofibrations, hence cofibrant in C'/sSet. Cobase change along $C' \to A'$ is a left Quillen functor $C'/\mathrm{sSet} \to A'/\mathrm{sSet}$; its right adjoint restricts along the structure map, which preserves fibrations and trivial fibrations. It preserves weak equivalences between cofibrant objects, so

$$A' \amalg_C B \cong A' \amalg_{C'} E \to A' \amalg_{C'} B'$$

is a weak equivalence. Also $A \to A \amalg_C B$ is a monomorphism (pushout of i), so left properness applied to a gives $A \amalg_C B \to A' \amalg_C B$ a weak equivalence. Composing the two proves the main claim.

For the homotopy-pushout assertion, factor $C \to A$ as a cofibration $C \hookrightarrow \widetilde{A}$ followed by a trivial fibration $\widetilde{A} \to A$. The span $\widetilde{A} \leftarrow C \hookrightarrow B$ is a cofibrant replacement of the given span: three cofibrant objects, both legs cofibrations. Its pushout is weakly equivalent to $A \amalg_C B$ by the main claim. □

The gluing lemma completes the foundations. Sections 3–6 build the relative prism and couniversality; Section 7 assembles the model structure.

3 The positive relative prism

The heart of the proof is a construction which turns a certain pair of poset maps into a strict Gray cylinder, with two crucial properties: the two endpoints are given by the two maps, and the cylinder is *stationary* on an old subset — at every parameter value it acts by the same prescribed functor. Stationarity is what allows the cylinder to survive pushouts: after an attachment identifies the old subset with something new, the whole cylinder lands inside the pushout.

The construction works at the level of directed chain complexes. A sieve $A \subseteq E$ of posets fixes a downward-closed subposet; on the new part $E \setminus A$ we contract to a constant. Then Steiner's functor ν converts the chain-level cylinder into a strict functor of ω-categories, and reflection τ_n carries it to finite n.

Throughout this section the interval is the right tensor factor. Write $\epsilon\colon D_1 \to D_0$ for the unique functor to the Gray unit, and set $\pi_B = \mathrm{id}_B \otimes \epsilon\colon B \otimes D_1 \to B$. For $i\colon A \to B$ and $v\colon A \to C$, a cylinder $P\colon B \otimes D_1 \to C$ is *stationary along i relative to v* if

$$P(i \otimes \mathrm{id}) = v\pi_A$$

as strict functors (including every cell of A).

3.1 The relative prism lemma

Lemma 3.1 (Relative prism)**.** *Let $A \subseteq E$ be a downward-closed subposet, and let $f, g\colon E \to E$ be monotone with*

$$f(e) \le g(e) \ \ (e \in E), \qquad f|_A = g|_A, \qquad g|_{E\setminus A} \text{ constant.}$$

For each $1 \le n \le \infty$ there is a strict functor

$$P_{f,g}\colon U_E \otimes D_1 \longrightarrow U_E$$

whose restrictions at the two interval endpoints are $U(f)$ and $U(g)$. Writing $i\colon A \hookrightarrow E$ and $v = fi = gi\colon A \to E$, one has the stationarity identity

$$P_{f,g}(U(i) \otimes \mathrm{id}_{D_1}) = U(v)\pi_{U_A}. \tag{3.1}$$

Consequently the endpoints induce simplicially homotopic maps on Street nerves, stationary relative to $N_nU(v)$.

Proof. Step 1: setup. If T is a nonempty finite totally ordered set, write $M[T]$ for the augmented normalized simplex chain complex on T with basis the strictly increasing tuples $[t_0, \dots, t_p]$, and with augmentation sending each vertex to 1. A weakly increasing tuple with a repeated entry means zero. The alternating deletion formula remains valid in the normalized quotient.

Fix a simplex $\sigma = (v_0 \le \cdots \le v_m)\colon \Delta[m] \to NE$ and put $T_\sigma = \{f(v_i), g(v_i) : 0 \le i \le m\} \subseteq E$. The indices with $v_i \in A$ form a prefix $0 \le i \le r$. On that prefix f and g coincide. If $r < m$, all remaining g-images equal a common $b = g(v_m)$, and $f(v_r) = g(v_r) \le f(v_{r+1}) \le b$. Hence T_σ is the ordered f-image list $[f(v_0), \dots, f(v_m)]$, possibly with b adjoined above. The order embedding $T_\sigma \hookrightarrow E$ gives $\nu M[T_\sigma] \to U_E^\infty$.

Step 2: the simplexwise chain map. Write e_0, e_1 for the vertices and u for the edge of $M[1]$, so $du = e_1 - e_0$. On the tensor basis of $M[m] \otimes M[1]$, prescribe for $x = [i_0 < \cdots < i_p]$, $w_j = v_{i_j}$:

$$\begin{aligned} C_\sigma(x \otimes e_0) &= F(w) := [f(w_0), \dots, f(w_p)], \\ C_\sigma(x \otimes e_1) &= G(w) := [g(w_0), \dots, g(w_p)], \\ C_\sigma(x \otimes u) &= h(w) := [f(w_0), \dots, f(w_p), g(w_p)]. \end{aligned} \tag{3.2}$$

All images are weakly increasing in T_σ; each is a positive basis element or zero. Both endpoint prescriptions are the normalized chain maps of the monotone maps f, g on vertices, hence preserve augmentation. So C_σ is augmentation-preserving in degree zero.

Step 3: chain identity. The only nontrivial check is on $x \otimes u$ with $p > 0$. We need $dh(w) = C_\sigma(d(x \otimes u))$. The left side is the alternating sum of the $p+2$ deletions of entries of $h(w)$; the right side equals

$$\sum_{j=0}^{p-1}(-1)^j h(w_0, \ldots, \widehat{w_j}, \ldots, w_p) + (-1)^p B' + (-1)^p(G(w) - F(w)), \tag{3.3}$$

with

$$B := [f(w_0), \ldots, f(w_{p-1}), g(w_p)], \qquad B' := [f(w_0), \ldots, f(w_{p-1}), g(w_{p-1})].$$

Note $h(w_0, \ldots, \widehat{w_p}) = B'$. The j-labelled terms for $j < p$ coincide on both sides. So it suffices to compare tails: $dh(w)$ has tail $(-1)^p B + (-1)^{p+1}F(w)$, while $C_\sigma(d(x \otimes u))$ has tail $(-1)^p B' + (-1)^p(G(w) - F(w))$. These agree iff $B - F(w) = B' + G(w) - F(w)$, i.e. $B - B' = G(w)$. We verify this in the two possible cases for w_{p-1}.

Case $w_{p-1} \in A$. Then $w_0, \ldots, w_{p-1} \in A$ (downward closed), so $f(w_j) = g(w_j)$ for $j < p$. If also $w_p \in A$, then $B = G(w)$, and B' has repeated last two entries, so $B' = 0$. If $w_p \notin A$, then $f(w_j) = g(w_j)$ for $j < p$ gives $B = G(w)$, and $g(w_{p-1}) = f(w_{p-1})$ makes the last two entries of B' equal, so $B' = 0$. Either way $B - B' = G(w)$.

Case $w_{p-1} \notin A$. Since A is a sieve and $w_{p-1} \leq w_p$, also $w_p \notin A$. Constancy of g on $E \setminus A$ gives $g(w_{p-1}) = g(w_p) = b$, so $B = B'$ and $G(w)$ has repeated last two entries, so $G(w) = 0$. Again $B - B' = G(w)$.

The case $p = 0$ reduces to $d[f(w_0), g(w_0)] = [g(w_0)] - [f(w_0)]$. This proves (3.2) is a positive augmentation-preserving chain map in every degree. The strong Steiner tensor identification (2.3) gives the strict functor

$$\mathcal{O}_m \otimes D_1 \cong \nu(M[m] \otimes M[1]) \xrightarrow{\nu C_\sigma} \nu M[T_\sigma] \longrightarrow U_E^\infty. \tag{3.4}$$

Step 4: naturality in the simplex. Let $\alpha\colon [q] \to [m]$ be monotone and $\rho = \sigma\alpha$. The embeddings of simplex values give $j\colon T_\rho \hookrightarrow T_\sigma$ as an ordered subset. On the tensor basis we claim

$$M(j)C_\rho = C_\sigma(M(\alpha) \otimes \mathrm{id}_{M[1]}). \tag{3.5}$$

Fix a basis tuple $[a_0 < \cdots < a_p]$ in $M[q]$. If α is injective on it, both sides equal the same tuple in (3.2) selected by the indices $\alpha(a_0), \ldots, \alpha(a_p)$, on tensoring with e_0, e_1, or u. If α is not injective on the tuple, then the right side is 0 by normalization. On the left, equal selected indices yield repeated f-images in the e_0 and u prescriptions, and repeated g-images in the e_1 prescription, so all three images are 0. These cases include faces, degeneracies, and every composite; naturality of the strong Steiner identification and functoriality of ν turn (3.5) into an equality of strict functors after mapping into U_E^∞.

Note: the vanishing of $h(w)$ as a *chain* does not collapse C_σ to identity on a trivial cell. When the top entry of a q-cell is zero (with $q > 0$), the two $(q-1)$-entries of its cell table coincide, so the induced cell is the identity on a common $(q-1)$-cell. That common cell need not itself be an identity. Equation (3.5) compares the whole chain map degree-by-degree, so it captures agreement on these retained boundaries as well.

Step 5: assembly. The categorification formula (2.1), together with cocontinuity of the Gray tensor in its first variable gives

$$U_E^\infty \otimes D_1 \cong \operatorname*{colim}_{\sigma\colon \Delta[m] \to NE} (\mathcal{O}_m \otimes D_1).$$

The compatible maps (3.4) consequently induce $P_{f,g}^\infty\colon U_E^\infty \otimes D_1 \to U_E^\infty$. The endpoint maps are $U^\infty(f)$ and $U^\infty(g)$ by their simplexwise formulas. The colimit is taken in strict ω-categories; tensor cocontinuity suffices.

Step 6: stationarity. On a simplex σ entirely in A, the maps F and G coincide and $h(w) = 0$ for every selected tuple. Precisely, the whole chain map C_σ equals $F \circ (\mathrm{id}_{M[m]} \otimes \epsilon)$, where ϵ is the interval augmentation ($\epsilon(e_0) = \epsilon(e_1) = 1$, $\epsilon(u) = 0$). Applying ν and the simplex colimit proves (3.1).

Step 7: descent to finite n. For $n < \infty$, let $q_E \colon U_E^\infty \to U_E$ be the reflection unit. By Lemma 2.1, $q_E P_{f,g}^\infty$ factors uniquely as

$$U_E^\infty \otimes D_1 \xrightarrow{q_E \otimes \mathrm{id}} U_E \otimes D_1 \xrightarrow{P_{f,g}} U_E.$$

Endpoint identities follow by precomposing with q_E and using the universal property of reflection. To verify (3.1) after reflection, precompose both sides with $q_A \otimes \mathrm{id}$; the resulting equality is the ω-case. Injectivity in the bijection of Lemma 2.1 gives the equality with target U_E. For $n = \infty$ no reflection is needed. Empty A or E uses the empty colimit, and the statements follow. Finally Lemma 2.3 gives the stated simplicial homotopy. □

3.2 Gluing a stationary cylinder through a pushout

The stationary cylinder extends through any pushout, by transposing to the right internal hom.

Lemma 3.2 (Gluing a stationary cylinder)**.** *Let $i \colon B_0 \to B$ and $a \colon B_0 \to X$ be arbitrary strict functors of n-categories, and form the pushout*

$$\begin{array}{ccc} B_0 & \xrightarrow{i} & B \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow[j]{} & Y. \end{array}$$

Suppose $F_0, F_1 \colon B \to B$ satisfy $F_0 i = F_1 i = i$, and suppose $P \colon B \otimes D_1 \to B$ has endpoints F_0, F_1 and satisfies $P(i \otimes \mathrm{id}) = i\pi_{B_0}$. Then there is a cylinder $Q \colon Y \otimes D_1 \to Y$ stationary on X, whose endpoints $\overline{F}_0, \overline{F}_1 \colon Y \to Y$ are specified by $\overline{F}_\delta j = j$ and $\overline{F}_\delta b = bF_\delta$ for $\delta = 0, 1$. Its restriction along $b \otimes \mathrm{id}$ is bP. This construction is natural in $a \colon B_0 \to X$, and its nerve gives a simplicial homotopy between $N_n\overline{F}_0$ and $N_n\overline{F}_1$ stationary along $N_n j$.

Proof. Let $H(Y) \in n\mathrm{Cat}$ represent functors $(-) \otimes D_1 \to Y$; this is an n-category by Lemma 2.1 (and automatic for $n = \infty$). The cylinders $bP \colon B \otimes D_1 \to Y$ and $j\pi_X \colon X \otimes D_1 \to Y$ transpose to maps $B \to H(Y)$ and $X \to H(Y)$. That these maps agree on B_0 is the identity

$$bi\pi_{B_0} = ja\pi_{B_0},$$

which follows from the stationarity of P and the defining relation $ji = ba$. The pushout universal property in $n\mathrm{Cat}$ gives a unique map $Y \to H(Y)$ extending both; its transpose is Q. The endpoint formulas and $Q(j \otimes \mathrm{id}) = j\pi_X$ follow from naturality of the adjunction and the same pushout universal property. The restriction to B is bP by construction. Naturality in a follows because for $X \to X'$ commuting with a the two maps $Y \to H(Y')$ agree on X and B. Lemma 2.3 gives the final homotopy assertion. No injectivity, freeness, or cofibrancy on i, a, X is used. □

4 Labelled pieces of algebraic pushouts

We fix $1 \le n \le \infty$. The labelled-piece formalism lets us compute the fibers of the nerve of a pushout of strict n-categories, and it lets us write a pushout of nerves as a union of two-piece covers, so that a weak-equivalence conclusion on the pieces glues to the whole.

The underlying idea is a bookkeeping device. Label every object of the target category by 0 or 1; the labelled formula identifies the 0-fiber and 1-fiber of the pushout with pushouts of the corresponding fibers of the legs. Two labels combine by a maximum or minimum of monotone maps, and the two-piece structure falls out automatically, even when the attaching functor identifies old objects or collapses old cells to identities.

A *full sub-n-category* of X on a set of objects S has exactly those objects and contains every positive-dimensional cell whose 0-source and 0-target lie in S. It is a *sieve* if a 1-cell $x \to y$ with $y \in S$ forces $x \in S$, and a *cosieve* if a 1-cell $x \to y$ with $x \in S$ forces $y \in S$. A sieve is equivalently the fiber over 0 of a strict functor $X \to D_1$, and a cosieve is the fiber over 1: the labelled object assignment extends uniquely to a strict functor because each nonempty hom in the target is terminal, and the sieve/cosieve condition excludes arrows labelled $1 \to 0$. Fibers so obtained are full, since all cells between objects with the same label must map to identities at that label.

4.1 Fibers preserve colimits

Lemma 4.1 (Fibers and objects preserve colimits)**.** *For $\epsilon \in \{0, 1\}$ the functor*

$$F_\epsilon \colon n\mathrm{Cat}/D_1 \longrightarrow n\mathrm{Cat}, \qquad (p \colon X \to D_1) \longmapsto X_\epsilon = p^{-1}(\epsilon),$$

preserves all small colimits. The object functor $\mathrm{Ob} \colon n\mathrm{Cat} \to \mathrm{Set}$ *also preserves all small colimits. Both hold for $n = \infty$.*

Proof. Build a right adjoint Q_0 to F_0. Given B, place all its objects over 0 and adjoin a new object $*$ over 1. Retain the homs $B(b, b')$, and set

$$Q_0(B)(b, *) = \mathbf{1}, \quad Q_0(B)(*, b) = \varnothing, \quad Q_0(B)(*, *) = \mathbf{1},$$

where $\mathbf{1}$ and $\varnothing$ are terminal and initial in the relevant dimension. All remaining compositions either have empty domain or terminal codomain, hence are uniquely determined; the strict n-category laws follow from those of B or by uniqueness. For $p \colon X \to D_1$ a functor $X_0 \to B$ extends uniquely over D_1 to $X \to Q_0(B)$: all objects of X_1 go to $*$, their internal cells and cells from X_0 to X_1 take unique images, and no cell goes from X_1 to X_0 since its 1-boundary would map to an arrow $1 \to 0$. So $\mathrm{Hom}_{n\mathrm{Cat}/D_1}(X, Q_0(B)) \cong \mathrm{Hom}_{n\mathrm{Cat}}(X_0, B)$ and F_0 is a left adjoint. The F_1 case uses a new initial object over 0 with terminal homs into B. Both constructions include $B = \varnothing$ and so preserve the empty colimit.

For Ob, let $K(S)$ be the ordinary chaotic category on S regarded as an n-category with identity cells above dimension one. Every $\mathrm{Ob}\, X \to S$ extends uniquely to $X \to K(S)$; hence Ob is a left adjoint. □

4.2 Oriental segments and poset pieces

Lemma 4.2 (Full oriental segments)**.** *Let I be an initial or final segment of $[m] = \{0, \dots, m\}$. If I is nonempty with $q + 1$ vertices, the face inclusion identifies $\tau_n \mathcal{O}_q$ with the full sub-n-category of $\tau_n \mathcal{O}_m$ on I. If $I = \varnothing$, that full subcategory is empty.*

Proof. The atoms of an oriental are indexed by strictly increasing tuples; the 0-source and 0-target of the atom at $[i_0 < \cdots < i_p]$ are i_0 and i_p. All cells are finite composites and identities of atoms, and every 1-cell has nondecreasing endpoints. Suppose a cell has 0-source a and 0-target b. Induction on the expression shows every occurring atom has all vertices in $[a, b]$: for an atom this is the tuple description, for a 0-composite the factors have endpoints (a, c) and (c, b) with $a \le c \le b$, for a

composition in positive dimension both factors keep the endpoints of the composite, and identities preserve the property. If $a, b \in I$, then $[a, b] \subseteq I$ and the expression uses only atoms of that face. So the face generates every cell on those objects.

For finite $n \geq 1$, reflection leaves objects unchanged, and every cell of dimension $\leq n$ is represented by a cell of the source. The same argument applied to a representative shows the reflection surjects onto the full subcategory. For injectivity, the inclusion $I \hookrightarrow [m]$ has a monotone retraction (clamp to the last vertex of an initial segment, or to the first of a final one), which by cosimplicial functoriality induces a retraction on orientals and then on reflections; the face map is injective on every cell set. The empty case has no objects and no cells. □

Lemma 4.3 (Poset pieces)**.** *There is a canonical identification* $\mathrm{Ob}\, U_P = P$ *for every poset* P*. If* $S \subseteq P$ *is a sieve, the map* $U_S \to U_P$ *is the inclusion of the full* 0*-fiber of the membership label* $U_P \to D_1$*. If* S *is a cosieve, it is the inclusion of the full* 1*-fiber. These identifications are natural for maps of labelled posets and include empty posets and empty fibers.*

Proof. Use the canonical simplex presentation

$$U_P = \operatorname*{colim}_{\sigma\colon \Delta[m] \to NP} \tau_n \mathcal{O}_m. \tag{4.1}$$

By Lemma 4.1 its object set is the colimit of vertex sets $[m]$, and sending the jth vertex of a simplex σ to $\sigma(j)$ identifies it with P: each element occurs as a zero-simplex, and face inclusions identify every vertex with that occurrence.

For a sieve S, the characteristic map $P \to [1]$ (0 on S, 1 on the complement) is monotone and induces a label on U_P, because $U_{[1]} = D_1$. The presentation (4.1) is a colimit in $n\mathrm{Cat}/D_1$ as well (slice colimits have underlying colimits in nCat). In a σ-piece the vertices mapping to S form an initial segment I_σ. By Lemma 4.2 the 0-fiber of that piece is its truncated segment oriental. Taking fibers of the colimit commutes with the colimit by Lemma 4.1.

The resulting colimit of segment orientals is canonically U_S: restricting simplex segments to S gives compatible maps to U_S; each simplex in S embeds fully; the composites are identities on simplices of S and on nonempty segments via the face inclusion selecting I_σ (a simplicial-set morphism in NP), while empty segments have nothing to check. The isomorphism obtained is the map $U_S \to U_P$ landing in its fiber. Fibers are full sieves; the cosieve case uses label 1 on S, final vertex segments, and F_1. Everything commutes with maps of labelled posets. □

4.3 Fibers of a labelled pushout

Proposition 4.4 (Labelled attachments)**.** *Let* $A \subseteq P$ *be a full subposet,* $a\colon U_A \to X$ *any strict functor, and form* $Y = X \amalg_{U_A} U_P$*. Suppose a monotone* $\lambda\colon P \to [1]$ *and a strict functor* $\xi\colon X \to D_1$ *satisfy* $\xi a = U(\lambda|_A)$*. Write* $P_\epsilon = \lambda^{-1}(\epsilon)$*,* $A_\epsilon = A \cap P_\epsilon$*,* $X_\epsilon = \xi^{-1}(\epsilon)$*. Then the induced label* $Y \to D_1$ *has actual full fibers*

$$Y_\epsilon \cong X_\epsilon \amalg_{U_{A_\epsilon}} U_{P_\epsilon}. \tag{4.2}$$

Independently of the labels,

$$\mathrm{Ob}\, Y \cong \mathrm{Ob}\, X \amalg_A P \cong \mathrm{Ob}\, X \amalg (P \setminus A). \tag{4.3}$$

For two compatible pairs (λ_i, ξ_i)*,* $i = 1, 2$*, with induced labels* $\bar{\lambda}_i$ *on* Y*, the* 0*-fibers intersect in the* 0*-fiber of* $\max \circ (\bar{\lambda}_1, \bar{\lambda}_2)$*, and the* 1*-fibers in the* 1*-fiber of* $\min \circ (\bar{\lambda}_1, \bar{\lambda}_2)$*. Formula* (4.2) *computes these intersections using the same max or min on both legs.*

Proof. Compatibility makes the defining pushout a pushout in $n\mathrm{Cat}/D_1$. Applying F_ϵ and then Lemmas 4.1 and 4.3 gives (4.2). The map from that pushout into Y identifies with the inclusion of a full fiber. The object formula follows from the object part of Lemma 4.1 and Lemma 4.3: its last isomorphism uses only the injectivity of $A \hookrightarrow P$, so old vertices of P are replaced by their images in X while vertices in $P \setminus A$ remain distinct. No injectivity of $\mathrm{Ob}\, a$ is required.

Two labels on Y give $Y \to D_1 \times D_1$. The monotone functions $\max, \min\colon [1] \times [1] \to [1]$ are strict functors, so their composites are labels. An object has maximum label 0 iff both labels are 0, and minimum label 1 iff both are 1. Full subcategories on the same objects coincide. The composite labels restrict to the corresponding max/min labels on both legs, so (4.2) applies to the intersections. □

Corollary 4.5 (Sieve and cosieve covers after attachment)**.** *With the notation above, the following are canonical identifications of subcategories of Y.*

(i) *If P_1, P_2 are sieves of P containing A and covering P, then $Y_i = X \amalg_{U_A} U_{P_i}$ $(i = 1, 2)$ are sieves covering Y on objects, and $Y_1 \cap Y_2 = X \amalg_{U_A} U_{P_1 \cap P_2}$.*

(ii) *If T_1, T_2 are cosieves of P covering P with $A \subseteq T_1$ and $A \cap T_2 = \varnothing$, then $Y_1 = X \amalg_{U_A} U_{T_1}$ and $Y_2 = U_{T_2}$ are cosieves covering Y on objects, and $Y_1 \cap Y_2 = U_{T_1 \cap T_2}$.*

If A is itself a sieve of P, then $X \to Y$ is the inclusion of a full sieve.

Proof. For (i), label membership in P_i by 0 and label every object of X by 0. Apply Proposition 4.4 to the two labels and their maximum. For (ii), label membership in each T_i by 1; use constant label 1 on X for T_1 and constant label 0 on X for T_2. These agree on the attaching source. The 1-fibers of the old-side labels are empty, and so is their minimum; formula (4.2) gives U_{T_2} and $U_{T_1 \cap T_2}$ exactly. In both cases (4.3) proves the object cover. When A is a sieve of P, label A and X by 0 and $P \setminus A$ by 1: the 0-fiber is $X \amalg_{U_A} U_A = X$. □

In particular, an attaching functor may identify old objects, collapse old cells to identities, or identify parallel old cells. None of these affect the cover calculation, since the labels agree on every old cell before the pushout is taken.

4.4 Nerves of two-piece covers

Lemma 4.6 (Two-piece covers)**.** *Let $C_1, C_2 \subseteq C$ be two full sieves (or two full cosieves) that cover the objects of C. Put $C_{12} = C_1 \cap C_2$. Then*

$$\begin{array}{ccc} N_n C_{12} & \hookrightarrow & N_n C_2 \\ \downarrow & & \downarrow \\ N_n C_1 & \hookrightarrow & N_n C \end{array}$$

is an ordinary pushout of simplicial sets. All four maps are monomorphisms, so it is a homotopy pushout.

Proof. A full inclusion gives a nerve monomorphism, and the intersection of the two nerves is $N_n C_{12}$: a functor whose image lies in both subcategories factorises uniquely through their full intersection. It remains to prove every simplex $z\colon \mathcal{O}_m \to C$ lies in one of the two nerves. For a sieve cover choose a piece containing $z(m)$; for $j \le m$ the oriental has an arrow $j \to m$ whose image lies in that piece; fullness then puts every cell of the simplex there. For a cosieve cover choose a piece containing $z(0)$ and use the arrows $0 \to j$. This handles $m = 0$ as well. So the union is the whole $N_n C$ degreewise; the intersection pushout holds because the two pieces are simplicial subsets with intersection the

third. Monomorphy of the legs is immediate, and the homotopy-pushout assertion follows from Lemma 2.5. □

The labelled-piece machinery is now complete. Section 5 uses it to glue local contractions into a couniversality statement for sieves with a right-adjoint retraction.

5 Couniversality and the unit on posets

Fix $n \in \{1, 2, \dots\} \cup \{\infty\}$. Call a strict functor a *Thomason equivalence* if its Street nerve is a weak homotopy equivalence. The term uses no model structure on $n\mathrm{Cat}$. We prove that the inclusion $A \hookrightarrow E$ of a sieve into a poset, when it has a right-adjoint retraction, remains a Thomason equivalence after every pushout. From this we deduce that the unit of $c_n \dashv N_n$ on every poset is a Thomason equivalence, independently of transfer.

A right-adjoint retraction of a poset inclusion $i\colon A \hookrightarrow E$ is a monotone $r\colon E \to A$ with $ri = \mathrm{id}_A$ and

$$a \le e \iff a \le r(e) \qquad (a \in A,\ e \in E). \tag{5.1}$$

We identify A with its image. In particular $r(e) \le e$, and A is a *sieve* when $e \le a \in A$ forces $e \in A$. For any attaching functor $a\colon U_A \to X$, write

$$Y = X \amalg_{U_A} U_E, \qquad j : X \to Y, \qquad v : U_E \to Y.$$

Below, elements of A are denoted b.

5.1 One greatest new element

Lemma 5.1 (Greatest new element)**.** *Let $i\colon A \hookrightarrow E$ be a sieve with a right-adjoint retraction, and suppose $E \setminus A$ has a greatest element. Then for every attaching functor a in $Y = X \amalg_{U_A} U_E$, the map $j\colon X \to Y$ admits a retraction $p\colon Y \to X$ with $N_n(jp)$ and id_{N_nY} joined by a zigzag of simplicial homotopies. Consequently j is a Thomason equivalence. Neither A nor E needs to be finite.*

Proof. Let x be greatest in $E \setminus A$ and $t = r(x) \in A$. If $b \in A$, $y \in E \setminus A$, $b \le y$, then $b \le y \le x$; by (5.1) we get $b \le t$. Define maps $E \to E$ by

$$k(e) = \begin{cases} e, & e \in A \\ t, & e \notin A \end{cases} \qquad g(e) = \begin{cases} e, & e \in A \\ x, & e \notin A \end{cases}.$$

Both are monotone: old-old and new-new cases go through unchanged; an old-new pair $e \le e'$ has $e \le t \le x$; a new-old pair cannot occur since A is a sieve. Thus

$$\mathrm{id}_E \le g, \qquad k \le g, \tag{5.2}$$

and all three restrict to id_A. The map g is constant on $E \setminus A$, so the relative prism (Lemma 3.1) applies to both (id_E, g) and (k, g).

Apply Lemma 3.2 to the two comparisons in (5.2). Let $G, K\colon Y \to Y$ be the pushout endomorphisms whose restrictions to X are j and whose restrictions to U_E are $vU(g)$ and $vU(k)$ respectively. The resulting right Gray cylinders are stationary on X; their endpoint pattern is

$$\mathrm{id}_Y \longrightarrow G \longleftarrow K.$$

By Lemma 2.3 this gives matching simplicial homotopies on Street nerves.

The image of k lies in A, so $k = i\bar{k}$ for some monotone $\bar{k}\colon E \to A$. The maps id_X and $aU(\bar{k})\colon U_E \to X$ agree on U_A, defining $p\colon Y \to X$ with $pj = \mathrm{id}_X$. Comparing restrictions to the two pushout summands gives $jp = K$: on X both are j; on U_E both are $vU(k) = vU(i)U(\bar{k}) = jaU(\bar{k})$. So the homotopy from id_Y to G followed by the reverse of the homotopy from jp to G exhibits $N_n p$ as a homotopy inverse of $N_n j$. The reversal is of an ordinary topological homotopy, not of a Gray cylinder. □

5.2 Small finite posets: induction through two-piece covers

Theorem 5.2 (Couniversality)**.** *Let $i\colon A \hookrightarrow E$ be a sieve of small posets with a right-adjoint retraction. Then $c_n N(i)$ is a couniversal Thomason equivalence: for every $a\colon U_A \to X$ the canonical map $X \to X \amalg_{U_A} U_E$ is a Thomason equivalence.*

Proof. Finite case. Induct on $|E \setminus A|$, simultaneously for all X and a. If $E \setminus A$ is empty, j is an isomorphism. If $E \setminus A$ has a greatest element, use Lemma 5.1.

Otherwise $E \setminus A$ has two distinct maximal elements x_1, x_2 (a finite poset with a unique maximal element has that element as its greatest). Each x_i is maximal in E: an element above a new element cannot be in the sieve A. Put

$$E_1 = E \setminus \{x_1\}, \quad E_2 = E \setminus \{x_2\}, \quad E_{12} = E \setminus \{x_1, x_2\}.$$

These are sieves of E containing A, with $E_1 \cup E_2 = E$ and $E_1 \cap E_2 = E_{12}$. The restriction of r to each E_T takes values in A and satisfies (5.1); each complement $E_T \setminus A$ is smaller. Put $Y_T = X \amalg_{U_A} U_{E_T}$, $j_T\colon X \to Y_T$. By induction each $N_n(j_T)$ is a weak homotopy equivalence.

Corollary 4.5(i), applied to the sieves E_1, E_2 of E, identifies $Y_i \to Y$ as full sieve inclusions; their intersection is Y_{12}; and they cover Y on objects. Lemma 4.6 then exhibits the bottom pushout of the commuting map of spans

$$\begin{array}{ccccc} N_nX & = & N_nX & = & N_nX \\ {\scriptstyle N_nj_1}\downarrow & & \downarrow{\scriptstyle N_nj_{12}} & & \downarrow{\scriptstyle N_nj_2} \\ N_nY_1 & \hookleftarrow & N_nY_{12} & \hookrightarrow & N_nY_2 \end{array}$$

as N_nY. The vertical maps are weak equivalences by induction; the horizontal legs are monomorphisms. Lemma 2.5 makes the induced map on pushouts a weak equivalence. The top pushout is N_nX and the bottom N_nY, so the map in question is N_nj. The finite induction is complete; no compatibility between local cylinders on different pieces is used.

Arbitrary posets. Write $q = ir\colon E \to E$; then $q^2 = q$. Let $\mathcal{F}$ be the collection of finite subsets $F \subseteq E$ with $q(F) \subseteq F$. It is filtered under inclusion (it contains $\varnothing$ and is closed under finite unions), and every finite $T \subseteq E$ is contained in the finite stable $T \cup q(T)$. Give each F the induced order and put $A_F = A \cap F$. Then A_F is a sieve of F and $r(F) \subseteq A_F$, so $r_F\colon F \to A_F$ is a right-adjoint retraction.

For each F put $Y_F = X \amalg_{U_{A_F}} U_F$, $j_F\colon X \to Y_F$. Each j_F is a Thomason equivalence by the finite case. Inclusions $F \subseteq F'$ give compatible transitions $Y_F \to Y_{F'}$ under X. Ordinary nerve simplices have finite vertex support, so

$$NE = \operatorname*{colim}_{F \in \mathcal{F}} NF, \qquad NA = \operatorname*{colim}_{F \in \mathcal{F}} NA_F.$$

The left adjoint c_n preserves colimits, so colimits commute:

$$\operatorname*{colim}_{F \in \mathcal{F}} Y_F \cong Y = X \amalg_{U_A} U_E.$$

Under this identification $\operatorname{colim} j_F = j$. By Lemma 2.4, N_n preserves these filtered colimits and weak homotopy equivalences are closed under filtered colimits. Hence $N_n j$ is a weak homotopy equivalence. The same proof applies verbatim to $n = \infty$. □

5.3 The unit on posets

The following is known [5, Cor. 10.11], [9, Thm. 6.9], [3, Prop. 9.2.3]; we include a prism-based proof.

Proposition 5.3 (The unit on every poset)**.** *For every small poset S the adjunction unit $\eta_S \colon NS \to N_n U_S$ is a weak homotopy equivalence. This is natural in S and holds for every finite $n \geq 1$ and for $n = \infty$.*

Proof. Finite case. Induct on $|S|$. For $S = \varnothing$ both sides are empty and η_S is an isomorphism. If S has a greatest element x, the pointwise inequality $\mathrm{id}_S \leq \mathrm{const}_x$ contracts NS. Lemma 3.1, with old sieve $\varnothing$, gives a right cylinder from id_{U_S} to $c_n N(\mathrm{const}_x)$, and the latter factors through $U_{\{x\}} = \mathcal{O}_0$, the terminal strict n-category. Lemma 2.3 contracts $N_n U_S$. Both sides are nonempty contractible, so η_S is a weak homotopy equivalence.

Otherwise choose two distinct maximal elements x_1, x_2 of S and put $S_i = S \setminus \{x_i\}$, $S_{12} = S \setminus \{x_1, x_2\}$. These are sieves, the two single deletions cover S, and all three posets are smaller. Their ordinary nerves cover NS: a simplex belongs to a sieve containing its last vertex. Lemmas 4.3 and 4.6 give a full sieve cover of U_S and of its Street nerve, with the joint zero label cutting the intersection out as $U_{S_{12}}$. Naturality of the unit gives a commuting map of spans

$$\begin{array}{ccccc} NS_1 & \hookleftarrow & NS_{12} & \hookrightarrow & NS_2 \\ {\scriptstyle \eta_{S_1}}\downarrow & & \downarrow{\scriptstyle \eta_{S_{12}}} & & \downarrow{\scriptstyle \eta_{S_2}} \\ N_n U_{S_1} & \hookleftarrow & N_n U_{S_{12}} & \hookrightarrow & N_n U_{S_2}. \end{array}$$

The vertical maps are weak equivalences by induction; the horizontal legs are monomorphisms. Lemma 2.5 makes the induced map on pushouts a weak equivalence. Under the cover identifications this is exactly η_S: on each NS_i it is that unit by naturality. This proves the finite case.

Arbitrary posets. Every poset is the filtered union of its finite full subposets. The ordinary nerve preserves this union since each simplex has finite support; c_n preserves the resulting colimit; N_n preserves the resulting filtered colimit by Lemma 2.4. Naturality identifies η_S with the filtered colimit of the units on the finite full subposets. Filtered-colimit closure of weak homotopy equivalences gives the result. Only the prism, cover, and gluing lemmas are used; the proof precedes and is independent of transfer. □

6 Comparison for subdivided attachments

We compare an ordinary pushout of Street nerves with the Street nerve of an actual attachment. Attaching functors are arbitrary; they may identify old objects and cells. The comparison supplies both the acyclicity of horn attachments and the preservation of weak equivalences under boundary attachments.

The strategy follows Thomason [16, Proposition 4.3]. Let $A \subseteq B$ be a sieve of posets whose inclusion into its upward closure Z has a right-adjoint retraction. Couniversality (Theorem 5.2) controls the attachment over Z; the two-piece cosieve cover with $B \setminus A$ then adds the rest of B. After proving this general comparison, we identify the twice-subdivided boundaries and horns as poset pairs with these properties.

6.1 The attachment comparison

Proposition 6.1 (Attachment comparison)**.** *Let $A \subseteq B$ be a sieve of small posets and Z its upward closure in B, with $A \hookrightarrow Z$ admitting a right-adjoint retraction. For every $a\colon U_A \to X$ in $n\mathrm{Cat}$, put*

$$V_B(X) = X \amalg_{U_A} U_B, \qquad P_B(X) = N_nX \amalg_{N_nU_A} N_nU_B.$$

Then

$$\gamma_{B,X}\colon P_B(X) \longrightarrow N_nV_B(X) \tag{6.1}$$

is a simplicial weak equivalence. The map $N_nU_A \to N_nU_B$ is a monomorphism. The comparison is natural in every $f\colon X \to X'$ under U_A.

Proof. Step 1: attachment over Z. Put $V_Z(X) = X \amalg_{U_A} U_Z$ and $P_Z(X) = N_nX \amalg_{N_nU_A} N_nU_Z$. Theorem 5.2 applied to $A \hookrightarrow Z$ gives $X \to V_Z(X) \in W_n$. Applying it with old target U_A gives $U_A \to U_Z \in W_n$. Since A is a sieve in Z, Lemma 4.3 identifies $U_A \to U_Z$ with an actual full inclusion; its Street nerve is monomorphic and a weak equivalence, hence a trivial cofibration. Its pushout $N_nX \to P_Z(X)$ is therefore a weak equivalence. The canonical triangle

$$\begin{array}{ccc} N_nX & \longrightarrow & P_Z(X) \\ & \searrow & \downarrow{\scriptstyle \gamma_{Z,X}} \\ & & N_nV_Z(X) \end{array}$$

then has $\gamma_{Z,X}$ a weak equivalence by two-out-of-three.

Step 2: adding the rest of B. Write $C = B \setminus A$ and $T = Z \setminus A = Z \cap C$. Both Z and C are cosieves in B and they cover B: an element outside C belongs to $A \subseteq Z$. Their intersection is T. Lemma 4.3 therefore gives cosieves $U_Z, U_C \subseteq U_B$ with intersection U_T. Apply Corollary 4.5(ii) with $P = B$, $T_1 = Z$, $T_2 = C$; its remaining hypotheses hold ($A \subseteq Z$, $A \cap C = \varnothing$). The canonical maps identify

$$V_Z(X),\ U_C \ \subseteq\ V_B(X), \qquad V_Z(X) \cap U_C = U_T \tag{6.2}$$

as full cosieves and their full intersection. These two cosieves cover all objects of $V_B(X)$; the corollary applies to the arbitrary functor a, including its identifications of old objects and cells.

Apply Lemma 4.6 to the two cosieve covers just obtained. It gives ordinary simplicial pushouts

$$N_nU_B = N_nU_Z \amalg_{N_nU_T} N_nU_C, \tag{6.3}$$

$$N_nV_B(X) = N_nV_Z(X) \amalg_{N_nU_T} N_nU_C. \tag{6.4}$$

All maps from the intersections to the pieces are monomorphisms. Reassociating the pushout defining $P_B(X)$ via (6.3) gives $P_B(X) = P_Z(X) \amalg_{N_nU_T} N_nU_C$. Under these identifications the comparison (6.1) is the map induced by $\gamma_{Z,X}$ and by $\mathrm{id}_{N_nU_C}$. In particular the square

$$\begin{array}{ccc} P_Z(X) & \hookrightarrow & P_B(X) \\ {\scriptstyle \gamma_{Z,X}}\downarrow & & \downarrow{\scriptstyle \gamma_{B,X}} \\ N_nV_Z(X) & \hookrightarrow & N_nV_B(X) \end{array}$$

is a pushout: $N_nV_Z(X) \amalg_{P_Z(X)} P_B(X)$ equals $N_nV_Z(X) \amalg_{N_nU_T} N_nU_C$. The top horizontal map is the cobase change of the specific monomorphism $N_nU_T \hookrightarrow N_nU_C$. Simplicial left properness, recorded

in Lemma 2.5, makes $\gamma_{B,X}$ a weak equivalence. Neither $N_nU_A \to N_nX$ nor $N_nU_T \to P_Z(X)$ needs to be monomorphic.

Step 3: monomorphy and naturality. That $N_nU_A \to N_nU_B$ is monic follows directly from the sieve identification in Lemma 4.3. Naturality: $\gamma_{B,X}$ is defined by canonical maps from N_nX and N_nU_B; for $f\colon X \to X'$ under U_A these commute with N_nf and the induced pushout functor $V_B(X) \to V_B(X')$, so the pushout universal property gives the strict square

$$\begin{array}{ccc} P_B(X) & \xrightarrow{\gamma_{B,X}} & N_nV_B(X) \\ \downarrow & & \downarrow \\ P_B(X') & \xrightarrow[\gamma_{B,X'}]{} & N_nV_B(X'). \end{array}$$

□

6.2 The twice-subdivided boundary and horn

Lemma 6.2 (The subdivision posets)**.** *Let $d\colon K \hookrightarrow \Delta[m]$ be a boundary inclusion ($m \geq 0$) or a horn inclusion $\Lambda^k[m] \hookrightarrow \Delta[m]$ ($m \geq 1$, $0 \leq k \leq m$). There exist finite posets $A \subseteq B$ and a canonical identification $\mathrm{Sd}^2 d = (NA \hookrightarrow NB)$. A is a sieve in B; its upward closure Z in B is a cosieve, and $A \hookrightarrow Z$ has a right-adjoint retraction.*

Proof. Put $M = \{0, \dots, m\}$, P the poset of nonempty subsets of M under inclusion. The subposet P_K of faces belonging to K is

$$P_K = \begin{cases} \{F : \varnothing \neq F \subsetneq M\}, & K = \partial\Delta[m], \\ \{F : \varnothing \neq F \subseteq M,\ M \setminus \{k\} \nsubseteq F\}, & K = \Lambda^k[m]. \end{cases}$$

For a horn, the excluded faces are M and $M \setminus \{k\}$, exactly those containing $M \setminus \{k\}$; P_K is downward closed.

Barycentric subdivision of an ordered simplicial complex has a vertex per nonempty face and a nondegenerate simplex per strict chain of faces. Hence $\mathrm{Sd}\Delta[m] = NP$ and $\mathrm{Sd}K = NP_K$; the identifications agree on every face and glue over K. The nondegenerate simplices of NP are nonempty chains in P with face given by deletion. Applying the description again gives

$$B = \{\sigma : \sigma \text{ a nonempty chain in } P\}, \qquad A = \{\sigma \in B : \sigma \subseteq P_K\}, \qquad \sigma \leq \tau \iff \sigma \subseteq \tau.$$

This proves the claimed identifications as simplicial sets, including degeneracies, and identifies the inclusion.

Every nonempty subchain of a chain in A again belongs to A, so A is a sieve of B. Its upward closure is $Z = \{\sigma \in B : \sigma \cap P_K \neq \varnothing\}$: for the nontrivial direction, a face $F \in \sigma \cap P_K$ gives $\{F\} \in A$ below σ; conversely a chain σ below some $\tau \in A$ must contain the members of τ, so $\sigma \cap P_K \neq \varnothing$. Define

$$r\colon Z \longrightarrow A, \qquad r(\sigma) = \sigma \cap P_K.$$

This is a nonempty chain (contains $\sigma \cap P_K$ as its own bottom chain), monotone, and fixes A. For $\alpha \in A$, $\sigma \in Z$: $\alpha \subseteq \sigma \iff \alpha \subseteq \sigma \cap P_K = r(\sigma)$, which is right adjunction.

For $m = 0$, $\partial\Delta[0] = \varnothing$, so B is a singleton, $A = Z = \varnothing$, and the adjunction is the identity of the empty poset. For a one-dimensional horn, $P_K = \{\{k\}\}$, so A is a singleton and Z consists of chains containing $\{k\}$. These cases are included in the formulas above; there are no zero-dimensional horn generators. □

Remark 6.3. For the zero-dimensional boundary $A = Z = T = \varnothing$ and $B = C$ is a singleton. So $V_Z(X) = X$ and $\gamma_{Z,X} = \mathrm{id}$; the two cosieves of $V_B(X) = X \amalg U_B$ are disjoint. The comparison is the isomorphism $N_n X \amalg \Delta[0] \cong N_n(X \amalg U_B)$.

6.3 Horn attachments are acyclic

Corollary 6.4 (Horn acyclicity)**.** *For every horn inclusion $d \in J$ and every attaching target, every pushout of $L_n d$ lies in W_n.*

Proof. Write $L_n d$ as $U_A \to U_B$ via Lemma 6.2. The nerve inclusion $NA \to NB$ is a weak equivalence: the horn inclusion is a simplicial weak equivalence, and subdivision preserves weak equivalences by two-out-of-three from the objectwise $\mathrm{Sd} \to \mathrm{id}$ [4, Theorem 4.5]. The adjunction unit gives a commuting square

$$\begin{array}{ccc} NA & \longrightarrow & NB \\ \downarrow & & \downarrow \\ N_n U_A & \longrightarrow & N_n U_B \end{array}$$

whose vertical maps are weak equivalences by Proposition 5.3. Two-out-of-three makes the lower map a weak equivalence; by Proposition 6.1 it is monic. So it is a trivial cofibration. For any attaching target $U_A \to X$ the pushout $N_n X \to P_B(X)$ is a weak equivalence. The composite

$$N_n X \longrightarrow P_B(X) \xrightarrow{\gamma_{B,X}} N_n V_B(X)$$

is the nerve of the attachment $X \to V_B(X)$, and both factors are weak equivalences; the attachment belongs to W_n. □

Corollary 6.5 (Invariance under attachment)**.** *Let $d \in I \cup J$, write $L_n d$ as $U_A \to U_B$ via Lemma 6.2, and let $f \colon X \to X' \in W_n$. For every $a \colon U_A \to X$, the induced functor*

$$X \amalg_{U_A} U_B \longrightarrow X' \amalg_{U_A} U_B$$

(the second attaching functor is fa) belongs to W_n.

Proof. $N_n X \to P_B(X)$ is monomorphic, being the cobase change of $N_n U_A \hookrightarrow N_n U_B$. There is a canonical pushout identification $P_B(X') = N_n X' \amalg_{N_n X} P_B(X)$. Simplicial left properness makes $P_B(X) \to P_B(X')$ a weak equivalence, since $N_n f$ is one. In the naturality square of Proposition 6.1, both horizontal maps are weak equivalences; two-out-of-three makes the right vertical map, the nerve of the asserted functor, a weak equivalence. □

The comparison and invariance leave the model-theoretic inputs in place. Section 7 assembles them into the model structure and the Quillen equivalence.

7 The model structure and the Quillen equivalence

Fix $n \in \{1, 2, \dots\} \cup \{\infty\}$. Corollary 6.4 supplies acyclic horn attachments; Corollary 6.5 supplies invariance under boundary attachment. We first extend the acyclicity through transfinite compositions, then build the model structure by the small object argument, prove properness, and finish the Quillen equivalence by checking the unit on an arbitrary simplicial set.

In plain terms, a model structure requires two weak factorization systems and compatibility of the three classes with each other. The small object argument builds each factorization system

one generating attachment at a time. The inputs from Section 6 are exactly what makes the horn attachments register as weak equivalences and what makes the boundary attachments preserve weak equivalences under cobase change. Without those two inputs the transferred classes would fail the model axioms.

7.1 The simplicial inputs

Write

$$L_n = c_n \mathrm{Sd}^2, \qquad R_n = \mathrm{Ex}^2 N_n, \qquad L_n \dashv R_n.$$

The Kan–Quillen model structure on sSet has weak equivalences W_{KQ}, fibrations $\mathrm{Fib}_{\mathrm{KQ}}$, and generating sets

$$I = \{\partial\Delta[m] \hookrightarrow \Delta[m] : m \geq 0\}, \qquad J = \{\Lambda^k[m] \hookrightarrow \Delta[m] : m \geq 1,\ 0 \leq k \leq m\}.$$

In particular $\partial\Delta[0] = \varnothing$.

The last-vertex transformation $\ell\colon \mathrm{Sd} \to \mathrm{id}$ is an objectwise weak equivalence [4, Theorem 4.5]; its mate $j\colon \mathrm{id} \to \mathrm{Ex}$ is also objectwise [10, Theorem III.4.6]. Both hold unconditionally. Naturality of j gives $j_{K'} f = (\mathrm{Ex} f) j_K$ for any $f\colon K \to K'$, so two-out-of-three shows that Ex preserves and reflects W_{KQ}. The same for ℓ and Sd. Hence the two descriptions of W_n in

$$W_n = N_n^{-1}(W_{\mathrm{KQ}}) = R_n^{-1}(W_{\mathrm{KQ}}), \qquad F_n = R_n^{-1}(\mathrm{Fib}_{\mathrm{KQ}}), \qquad C_n = {}^{\perp}(F_n \cap W_n)$$

agree. The class W_n has two-out-of-three and is closed under retracts. By Lemma 2.4 it is also closed under filtered colimits of arrows, without any injectivity of transitions.

For a set $\mathcal{S}$ of maps, write $\mathcal{S}$-inj for the maps with the right lifting property against $\mathcal{S}$, and $\mathcal{S}$-cof $= {}^{\perp}(\mathcal{S}\text{-inj})$. A relative $\mathcal{S}$-cell map is a transfinite composite of pushouts of coproducts of members of $\mathcal{S}$.

7.2 Transfinite horn attachments

Lemma 7.1 (Transfinite horn attachments)**.** *Every relative L_nJ-cell map lies in W_n.*

Proof. Corollary 6.4 proves one-attachment acyclicity with arbitrary target. A set of simultaneous attachments may be well ordered: at each stage map into the original target followed by the inclusion of the current stage. The universal property of the resulting colimit is the pushout of the coproduct. It suffices to handle a continuous ordinal sequence $(X_\alpha)_{\alpha \leq \lambda}$ in which each successor is a single attachment.

By transfinite induction $X_0 \to X_\alpha \in W_n$ for every α. Base: identity. Successor: compose with the weak equivalence from Corollary 6.4. Nonzero limit β: the ordinal of stages $\alpha < \beta$ is filtered; apply Lemma 2.4 to the diagram of arrows $X_0 \to X_\alpha$. Its colimit is $X_0 \to \mathrm{colim}_{\alpha<\beta} X_\alpha = X_\beta$ (the colimit of the constant source diagram is X_0). □

7.3 The model structure

Proposition 7.2 (Transferred model structure)**.** *The classes (C_n, W_n, F_n) form a combinatorial model structure on nCat, cofibrantly generated by (L_nI, L_nJ).*

Proof. The underlying category is locally presentable, hence complete and cocomplete, by Lemma 2.4. Choose an infinite regular cardinal κ above the presentability bound of every domain in $\mathcal{S} = L_nI$ or L_nJ (exists by local presentability, since $\mathcal{S}$ is a set).

The small-object argument. Given $X \to Y$, set $X_0 = X$. At stage $\alpha + 1$ attach to X_α one copy of the codomain of each member of $\mathcal{S}$ for every commutative square from that member to $X_\alpha \to Y$; this is a set of squares (local smallness). Use colimits at limit stages. The result is $X \xrightarrow{i} X_\kappa \xrightarrow{p} Y$. Here i is a relative $\mathcal{S}$-cell map and $p \in \mathcal{S}$-inj: any map from a generating domain to X_κ factors through some X_α (κ is κ-filtered), and its lifting square was attached at stage $\alpha + 1$, supplying the lift.

Pushouts, transfinite composites, and retracts of maps with a fixed left lifting property retain that property, so every relative $\mathcal{S}$-cell map lies in $\mathcal{S}$-cof. Conversely if $u \in \mathcal{S}$-cof, factor as $A \xrightarrow{i} Z \xrightarrow{p} B$ by the construction above; the square with top i, left u, right p, bottom id_B has a diagonal $s\colon B \to Z$. This gives $su = i$, $ps = \mathrm{id}_B$. Thus u is a retract of i in the arrow category with fixed source. Hence

$$(\mathcal{S}\text{-cof}, \mathcal{S}\text{-inj})$$

is a weak factorization system; its left class is exactly the retracts of relative $\mathcal{S}$-cell maps.

Lifting characterizations. The adjunction identifies lifting squares and their fillers, so

$$(L_nI)\text{-inj} = R_n^{-1}(I\text{-inj}) = F_n \cap W_n, \qquad (L_nJ)\text{-inj} = R_n^{-1}(J\text{-inj}) = F_n.$$

(Here I-inj in sSet is trivial Kan fibration $= W_{\mathrm{KQ}} \cap \mathrm{Fib}_{\mathrm{KQ}}$; J-inj is $\mathrm{Fib}_{\mathrm{KQ}}$.) Thus $(C_n, F_n \cap W_n)$ is a weak factorization system.

Second factorization system. Put $\mathcal{D} = (L_nJ)$-cof. Because $F_n \cap W_n \subseteq F_n$, we get $\mathcal{D} \subseteq C_n$. Lemma 7.1 and retract closure also give $\mathcal{D} \subseteq W_n$. For the reverse inclusion, take $u\colon A \to B \in C_n \cap W_n$ and factor by the second argument as $A \xrightarrow{i} Z \xrightarrow{p} B$ with $i \in \mathcal{D}$, $p \in F_n$. Then $i \in W_n$ and $u = pi \in W_n$, so $p \in F_n \cap W_n$ by two-out-of-three. Lifting of u against p supplies $s\colon B \to Z$ with $su = i$, $ps = \mathrm{id}_B$. The fixed-source retract puts $u \in \mathcal{D}$. Hence $\mathcal{D} = C_n \cap W_n$, so $(C_n \cap W_n, F_n)$ is a weak factorization system. Model axioms are complete: two-out-of-three and retract for W_n, and both weak factorization systems, on a bicomplete category; local presentability and the displayed generating sets make it combinatorial. □

7.4 Properness

Proposition 7.3 (Properness)**.** *The model structure of Proposition 7.2 is left and right proper.*

Proof. Right properness. Let $p\colon Y \to Z \in F_n$ and $w\colon Z' \to Z \in W_n$. The right adjoint R_n carries the pullback $Y \times_Z Z'$ to the simplicial pullback $R_nY \times_{R_nZ} R_nZ'$ (right adjoints commute with limits). Its map to R_nY is a weak equivalence by right properness of the Kan–Quillen structure, since R_np is a Kan fibration and R_nw a weak equivalence. So $Y \times_Z Z' \to Y \in W_n$.

Left properness. First take $i\colon X \to Y$ a relative L_nI-cell map and $w\colon X \to X' \in W_n$. Write i as a continuous sequence of individual boundary attachments X_α starting at X, and put $X'_\alpha = X' \amalg_X X_\alpha$. Inductively, $X_\alpha \to X'_\alpha \in W_n$: base is w; at a successor stage the second attachment has attaching map followed by $X_\alpha \to X'_\alpha$, so Corollary 6.5 proves the step; at a nonzero limit, Lemma 2.4 gives the statement. So $Y \to X' \amalg_X Y \in W_n$.

For general $i \in C_n$: the first small object factorization and its lifting retract (from the proof of Proposition 7.2) give i a retract of a relative L_nI-cell $j\colon X \to Z$ with the same source: $s\colon Y \to Z$, $p\colon Z \to Y$, $si = j$, $pj = i$, $ps = \mathrm{id}_Y$. Cobase change along w gives matching s', p' between $X' \amalg_X Y$ and $X' \amalg_X Z$ with $p's' = \mathrm{id}$. So $Y \to X' \amalg_X Y$ is a retract of $Z \to X' \amalg_X Z$, which is in W_n by the previous paragraph. Both properness arguments apply to arbitrary objects; no cofibrancy or fibrancy is used. □

7.5 The unit after two subdivisions

Let $\eta_T\colon T \to N_n c_n T$ denote the unit of $c_n \dashv N_n$. The poset case is Proposition 5.3. The next proposition extends to all simplicial sets.

Proposition 7.4 (The subdivided unit)**.** *For every simplicial set K, the map*

$$\theta_K = \eta_{\mathrm{Sd}^2 K}\colon \mathrm{Sd}^2 K \longrightarrow N_n L_n K \tag{7.1}$$

is a weak equivalence.

Proof. For $K = \Delta[m]$ and $K = \partial\Delta[m]$, Lemma 6.2 identifies $\mathrm{Sd}^2 K$ with the ordinary nerve of a poset; the assertion is Proposition 5.3. For $K = \varnothing$, both sides are empty.

Every simplicial set is obtained from $\varnothing$ by a continuous transfinite sequence of boundary attachments: order nondegenerate simplices by dimension and well-order those in each dimension; the boundary of a simplex maps into the already-constructed stage. The attaching map may identify vertices or faces, or send faces to degenerate simplices; the construction still recovers K by the unique expression of each simplex as a degeneration of a nondegenerate one.

Consider a successor step $K_+ = K_0 \amalg_{\partial\Delta[m]} \Delta[m]$ and assume the assertion for K_0. Put

$$V = \mathrm{Sd}^2 K_0, \quad A = \mathrm{Sd}^2 \partial\Delta[m], \quad B = \mathrm{Sd}^2 \Delta[m].$$

Since subdivision is a left adjoint, $\mathrm{Sd}^2 K_+ = V \amalg_A B$. Naturality of η gives a map

$$q\colon V \amalg_A B \longrightarrow N_n c_n V \amalg_{N_n c_n A} N_n c_n B. \tag{7.2}$$

The three maps on objects are weak equivalences: on V by induction, on A, B by the simplex and boundary cases. The leg $A \to B$ is monomorphic. Writing $A = NE$, $B = NF$ as in Lemma 6.2, the inclusion $E \subseteq F$ is a sieve; Lemma 4.3 makes $c_n A \to c_n B$ a sieve inclusion, so $N_n c_n A \to N_n c_n B$ is monomorphic. Lemma 2.5 with these two monomorphic legs makes q a weak equivalence; no monomorphism of $A \to V$ is required.

Apply Proposition 6.1 to the arbitrary attaching functor $c_n A \to c_n V$: the map

$$t\colon N_n c_n V \amalg_{N_n c_n A} N_n c_n B \longrightarrow N_n(c_n V \amalg_{c_n A} c_n B) \tag{7.3}$$

is a weak equivalence. Since c_n preserves colimits, its target is $N_n c_n(V \amalg_A B)$. The composite tq is exactly $\eta_{V \amalg_A B}$: on each pushout leg this is naturality of η, and equality on the two legs determines the map from the pushout. So θ_{K_+} is a weak equivalence, and noninjective boundary maps cause no change.

At a nonzero limit, Sd^2 and L_n preserve the colimit as left adjoints, and N_n preserves the filtered colimit by Lemma 2.4. The unit at that stage is the filtered colimit of previous units, hence a weak equivalence by the same lemma. Transfinite induction proves (7.1) for every K. Only the simplex and boundary descriptions of subdivision enter, so the argument is not affected by arbitrary attaching maps. □

7.6 The Quillen equivalence

Theorem 7.5 (Quillen equivalence)**.** *The adjunction*

$$L_n\colon \mathrm{sSet}_{\mathrm{KQ}} \rightleftarrows (n\mathrm{Cat}, C_n, W_n, F_n)\colon R_n$$

is a Quillen equivalence.

Proof. The lifting characterization from the proof of Proposition 7.2 shows that R_n preserves fibrations and trivial fibrations, so this is a Quillen adjunction.

Sub-calculation for the composite subdivision. Put $S = \mathrm{Sd}$, $T = \mathrm{Ex}$, and let $a_K \colon K \to TSK$ be the unit of $S \dashv T$. The mate identity is $j_K = T(\ell_K)a_K \colon K \to TK$. Since j_K is a weak equivalence, ℓ_K is, and T preserves weak equivalences, $T(\ell_K)$ is a weak equivalence; two-out-of-three gives a_K a weak equivalence. In particular a_{SK} is, and the unit b_K of $S^2 \dashv T^2$,

$$b_K = T(a_{SK})a_K \colon K \to T^2S^2K,$$

is a weak equivalence (because T preserves).

The composite unit. The unit of $L_n \dashv R_n$ is

$$u_K = T^2(\theta_K)\, b_K \colon K \to T^2N_nc_nS^2K = R_nL_nK.$$

Proposition 7.4 makes θ_K a weak equivalence, and T preserves; so u_K is a weak equivalence for every K. For any strict functor $f \colon L_nK \to Y$, its simplicial adjunct is $f^\sharp = R_n(f)u_K \colon K \to R_nY$. By definition of W_n and two-out-of-three,

$$f \in W_n \iff R_n(f) \in W_{\mathrm{KQ}} \iff f^\sharp \in W_{\mathrm{KQ}}.$$

This is the Quillen-equivalence criterion (for cofibrant K and fibrant Y). Together with Propositions 7.2 and 7.3, this proves every assertion of Theorem 1.1 for each finite $n \geq 1$ and for $n = \infty$. □

A Local finite presentability of $n\mathrm{Cat}$

We prove the presentability assertion used in Lemma 2.4. The subtle point is that an equation between boundary cells can create new composable pairs, so a presentation must generate *defined terms* and *equations* together. Taking a plain quotient of existing cells would miss the composites that become defined after the identification. All sets and constructions below lie in the fixed universe.

A.1 Signature and equations

Fix $n \in \{1, 2, \ldots, \infty\}$, with ∞ meaning ω. Use one sort C_d for each $0 \leq d \leq n$, or one per nonnegative integer when $n = \infty$. The total operations are

$$s_d, t_d \colon C_d \longrightarrow C_{d-1} \quad (d > 0), \qquad e_d \colon C_d \longrightarrow C_{d+1} \quad (d < n).$$

For $0 \leq k < d$, s_k^d, t_k^d, e_k^d are iterated. There is a binary partial operation $*_k^d$ on C_d with domain

$$s_k^d(x) = t_k^d(y). \tag{A.1}$$

We read $x *_k^d y$ as "x after y" and suppress superscripts when the sort is clear.

The equations are the strict-globular-category laws.

Globularity and identity boundary:

$$s_{d-1}s_d = s_{d-1}t_d, \quad t_{d-1}s_d = t_{d-1}t_d, \quad s_{d+1}e_d = \mathrm{id} = t_{d+1}e_d,$$

for $2 \leq d \leq n$ and $0 \leq d < n$, all d when $n = \infty$. For a pair satisfying (A.1), the composition boundary axioms are

$$\begin{aligned} &s_d(x *_{d-1} y) = s_d(y), && t_d(x *_{d-1} y) = t_d(x), \\ &s_d(x *_k y) = s_d(x) *_k s_d(y), && t_d(x *_k y) = t_d(x) *_k t_d(y) \quad (k < d-1). \end{aligned}$$

For each k, associativity is imposed on triples with $s_k(x) = t_k(y)$, $s_k(y) = t_k(z)$. Units are $x *_k e_k^d(s_k^d x) = x$ and $e_k^d(t_k^d x) *_k x = x$. Identities preserve every lower composition: $e_d(x *_k^d y) = e_d(x) *_k^{d+1} e_d(y)$.

Interchange, for $k < l < d$, is

$$(x *_l y) *_k (z *_l w) = (x *_k z) *_l (y *_k w), \tag{A.2}$$

guarded by all four inner matchings

$$s_l^d(x) = t_l^d(y), \quad s_l^d(z) = t_l^d(w), \quad s_k^d(x) = t_k^d(z), \quad s_k^d(y) = t_k^d(w).$$

These imply the outer comparisons by globularity; for example $s_k^d(x *_l y) = s_k^d(x) = s_k^d(y)$ and $s_k^d(z *_l w) = s_k^d(z) = s_k^d(w)$, so the left side is composable. Thus the guard is an ordinary composable-array condition and not an extra restriction. Definedness of the right-hand side alone does not supply the guard.

This is a set of conditional equations with finite terms and finitely many premises; no operation or equation takes infinitely many arguments. Its models are precisely strict globular n-categories. For finite n, adjoining iterated identities above dimension n gives the convention of Section 2.

A.2 Defined terms and presentations

A *sorted set of generators* G assigns a dimension to each generator. Let $T(G)$ be the set of all finite well-sorted raw terms. A raw term belongs to $T(G)$ even if its boundary equation is not yet known; it becomes a cell only once declared defined. Let R be a set of equations between raw terms of the same sort. An equation in R is *strong*: its interpretation requires both terms defined AND equal. In particular $t = t$ can be used to force t to be defined.

Lemma A.1 (Strict-category presentations)**.** *For every sorted G and every set R of strong equations, there is a small strict n-category $P(G, R)$ such that, naturally in Y, strict functors $P(G, R) \to Y$ correspond to sorted assignments of the generators to cells of Y satisfying R.*

Proof. Generate simultaneous judgments $\mathsf{D}(t)$ (defined) and $t = u$ (both defined, equal) by these rules.

(i) Every generator is defined and each member of R is an equation.

(ii) An equation $t = u$ implies $\mathsf{D}(t)$, $\mathsf{D}(u)$. Definedness of an operation term implies definedness of all arguments. Definedness of $x *_k y$ also gives the matching (A.1).

(iii) Total operations applied to defined arguments yield defined terms. Defined arguments satisfying (A.1) yield a defined composite.

(iv) Equality is symmetric and transitive. Reflexivity is $\mathsf{D}(t) \Rightarrow t = t$; it is not imposed on undefined raw terms.

(v) Equal arguments transport definedness and equality through an operation: from $t_i = u_i$ and $\mathsf{D}(f(t_i))$ infer $\mathsf{D}(f(u_i))$ and equality of the two operation terms.

(vi) Every ground instance of the strict-category equations above is imposed when its arguments are defined and its stated matchings hold; the conclusion is equality of the two sides, hence definedness of both. Interchange retains all four conditions in (A.2).

Globularity, identity boundary and unit equations need definedness of their variables; composition boundary and identity preservation from identities require (A.1); associativity requires its two matchings. Each rule has finitely many premises, and every rule is sound in a strict category.

Start with the assumptions and repeatedly add conclusions of all applicable rules. The union of finite stages is the least closed set of judgments; any rule's finitely many premises occur together at some stage; every resulting judgment has a finite derivation. Everything involved is a set.

Let $P(G,R)_d$ be defined terms of sort d modulo derivable equality. Equality rules make this an equivalence relation; rules (iii) and (v) descend total operations. For a partial operation, suppose the boundary classes of $[x]$, $[y]$ match. Total boundary terms are defined, so class equality means $s_k(x) = t_k(y)$ is derivable; rule (iii) declares $x *_k y$ defined. This is independent of representatives: $x = x'$ and $y = y'$ give $s_k(x) = s_k(x')$, $t_k(y) = t_k(y')$, so the matching descends and rule (v) carries the composites. Rule (ii) ensures matching boundaries on the quotient. Rule (vi) gives all strict category laws.

Now an assignment $G \to Y$ satisfying R. Induction on derivations shows every defined term evaluates in Y and every derived equation has equal values; rule (v) transports because equal arguments have identical boundary values, so existence of one partial value entails existence of the other. Evaluation therefore descends to a strict functor, unique by induction on terms. Conversely, every strict functor gives such an assignment. The construction is small, consistent (the trivial strict category satisfies everything), and empty when $G = \varnothing$ (no raw ground terms since all operations have positive arity). □

Remark A.2. The simultaneous construction is essential even for ordinary categories. If the source and target of a free arrow are identified, its presentation must create the square of the arrow and all higher powers; these are not quotients of the original cells, but new cells introduced by rule (iii) as soon as their boundaries become derivable.

A.3 Colimits, filtered colimits, finite presentations

Proposition A.3 (Local finite presentability)**.** *For every $1 \le n \le \infty$, the category n*Cat *is locally small, complete, cocomplete, and locally finitely presentable. Filtered colimits are computed on cells. Every set of objects has a common infinite regular presentability bound.*

Proof. Colimits. For a small diagram $X\colon I \to n\mathrm{Cat}$, take a generator $[i,a]$ per cell a of each X_i and impose the operation-table equations

$$s([i,a]) = [i,s(a)], \quad t([i,a]) = [i,t(a)], \quad e([i,a]) = [i,e(a)], \quad [i,a] *_k [i,b] = [i,a *_k b]$$

(the last whenever $a *_k b$ is defined), and, for each arrow $f\colon i \to j$, the equation $[i,a] = [j,X(f)(a)]$. By Lemma A.1, maps from this presentation to Y are exactly compatible families $X_i \to Y$; it is the colimit.

Limits and local smallness. In each dimension take the set of compatible families; source, target, identity descend coordinatewise, and composition is defined coordinatewise on exactly the matching pairs. All laws and the limit universal property hold coordinatewise, including the empty limit. Strict functors form a subset of the product of cell maps, so the category is locally small.

Filtered colimits. Form the colimit of each set of cells. Finitely many representatives occur at one stage; any matching equation holds at a later common stage where the composite is defined; a further common stage proves independence of choices. Each category law holds at a stage containing its finite tuple and matchings. No injectivity of transitions is used.

Finite-term witness. If finitely many assigned generator values make a finite collection of terms defined and satisfy specified equations in the colimit, those facts occur at one stage. Induct on subterms: lift arguments, lift the boundary equality needed for a partial composite, lift the composite, lift the required equalities. Each step uses the equality criterion for filtered colimits of sets.

Finite presentations. For finite G, R, a map $P(G,R) \to \mathrm{colim}_i X_i$ is an assignment of finitely many generator values satisfying finitely many strong equations, so it lifts to a stage. If two stage

maps are equal in the colimit, their values on G agree at a later common stage; by the presentation universal property the whole functors agree there. So

$$\operatorname*{colim}_i \operatorname{Hom}(P(G,R), X_i) \cong \operatorname{Hom}(P(G,R), \operatorname*{colim}_i X_i),$$

and $P(G,R)$ is finitely presentable.

A generating set. Every strict category X has its full *cell presentation*: one generator g_a per cell and all operation-table equations. Evaluation is onto and, by induction on defined terms together with the table equations, injective. Write it (G_X, R_X). Consider the finite pairs (G_0, R_0) with $G_0 \subseteq G_X$, $R_0 \subseteq R_X$, every generator in R_0 contained in G_0. Ordered by inclusion, these form a nonempty filtered poset (empty pair allowed, finite unions upper). Their presented categories have colimit $P(G_X, R_X)$: a compatible family of maps from finite presentations assigns a value to every generator and enforces every relation. The finite pairs need not be closed under operations; finite presentations themselves supply closures. Thus every object is a filtered colimit of objects from a set of finitely presentable ones. With cocompleteness, nCat is locally finitely presentable. The same argument goes through with the countable signature in the ω case.

Common regular bound. For a cardinal κ, the same proof makes $P(G,R)$ κ-presentable when $|G|, |R| < \kappa$ and κ is infinite regular: fewer than κ generator values and fewer than κ relations must be lifted, and the witnesses form a diagram of size $< \kappa$ so a cocone places them at one stage. Equality of maps is detected by fewer than κ generator equalities. The full cell presentation of X has cardinality at most $\max\{\aleph_0, |\coprod_d X_d|\}$; choose a larger infinite regular cardinal. For a set of objects, bound their full presentations and choose one larger infinite regular cardinal. □

References

[1] J. Adámek, M. Hébert, and J. Rosický. On essentially algebraic theories and their generalizations. *Algebra Universalis*, 41(3):213–227, 1999.

[2] J. Adámek and J. Rosický. *Locally Presentable and Accessible Categories*, volume 189 of *LMS Lecture Note Series*. Cambridge University Press, 1994.

[3] D. Ara. Théorie de l'homotopie des ∞-catégories strictes. Habilitation à diriger des recherches, 2023. `https://www.normalesup.org/~ara/files/hdr.pdf`.

[4] D. Ara and G. Maltsiniotis. Vers une structure de catégorie de modèles à la Thomason sur la catégorie des n-catégories strictes. *Advances in Mathematics*, 259:557–654, 2014. arXiv:1305.5086v3.

[5] D. Ara and G. Maltsiniotis. Le type d'homotopie de la ∞-catégorie associée à un complexe simplicial. arXiv:1503.02720v1, 2015.

[6] D. Ara and G. Maltsiniotis. *Joint et tranches pour les ∞-catégories strictes*, volume 165 of *Mémoires de la SMF*. SMF, 2020. arXiv:1607.00668v4; DOI 10.24033/msmf.473.

[7] D.-C. Cisinski. La classe des morphismes de Dwyer n'est pas stable par rétractes. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 40(3):227–231, 1999.

[8] T. M. Fiore and S. Paoli. A Thomason model structure on the category of small n-fold categories. *Algebraic & Geometric Topology*, 10(4):1933–2008, 2010.

[9] A. Gagna. Strict n-categories and augmented directed complexes model homotopy types. *Advances in Mathematics*, 331:542–564, 2018. arXiv:1612.04450v2.

[10] P. G. Goerss and J. F. Jardine. *Simplicial Homotopy Theory*, volume 174 of *Progress in Mathematics*. Birkhäuser, Basel, 1999.

[11] L. Guetta and G. Maltsiniotis. Homologie polygraphique des systèmes locaux. *Advances in Mathematics*, 448:109719, 2024. arXiv:2309.10466v2.

[12] M. Hébert, J. Adámek, and J. Rosický. More on orthogonality in locally presentable categories. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 42(1):51–80, 2001.

[13] R. Steiner. ω-categories and chain complexes. *Homology, Homotopy and Applications*, 6(1):175–200, 2004. arXiv:math/0403237v2.

[14] R. Steiner. Orientals. In *Categories in Algebra, Geometry and Mathematical Physics*, volume 431 of *Contemp. Math.*, pages 427–439. AMS, 2007. arXiv:math/0601383v2.

[15] R. Street. The algebra of oriented simplexes. *Journal of Pure and Applied Algebra*, 49(3):283–335, 1987.

[16] R. W. Thomason. *Cat* as a closed model category. *Cahiers de Topologie et Géométrie Différentielle*, 21(3):305–324, 1980.

[17] K. Worytkiewicz, K. Hess, P.-E. Parent, and A. Tonks. A model structure à la Thomason on *2-Cat*. *Journal of Pure and Applied Algebra*, 208(1):205–236, 2007.

[18] K. Worytkiewicz, K. Hess, P.-E. Parent, and A. Tonks. Corrigendum to "A model structure à la Thomason on *2-Cat*". *Journal of Pure and Applied Algebra*, 220(12):4017–4023, 2016.